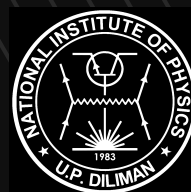


General Relativity I

Notes

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Chapter 1

Special Relativity

We first review a few concepts from Special Relativity.

1.1 Inertial Frames and the Principle of Relativity

Newton's first law of motion reads

An object at rest remains at rest, and an object in motion remains in motion at constant speed and in a straight line unless acted on by an unbalanced force.

An *inertial* reference frame can be defined as one in which this law holds. Thus, in the absence of unbalanced forces, the spatial coordinates (x, y, z) of any point particle, as viewed in the reference frame S with coordinates (ct, x, y, z) , must satisfy the condition that says that the particle cannot accelerate:

$$\frac{d^2x}{dt^2} = \frac{d^2y}{dt^2} = \frac{d^2z}{dt^2} = 0. \quad (1.1)$$

The space and time coordinates (spacetime coordinates) of any two inertial reference frames S and S' must, therefore, be related by a linear transformation. Otherwise, if the above equations are satisfied by the spacetime coordinates of the particle as measured in the reference frame S , then they would *not* be satisfied by corresponding spacetime coordinates in the reference frame S' !

For simplicity, let us suppose that S' is moving with respect to S with speed v in the $+x$ direction, that their coordinate axes are aligned, and that their origins coincide at $t = t' = 0$ (c.f. Figure 1.1 on page 2 of the textbook). Then, obviously, nothing happens to the spatial directions transverse to x , namely y and z . In other words, $y' = y$ and $z' = z$. All the interesting stuff happens in the t - x plane:

$$ct' = Act + Bx \quad (1.2a)$$

$$x' = Cct + Dx. \quad (1.2b)$$

We can write this as a vector equation:

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \quad (1.3)$$

We need $x' = 0$ to correspond to the line $x = vt$ since if the particle is moving with the same velocity as S' as measured in S , then the particle must not be moving at all as measured in the frame S' . But, $x' = 0$ corresponds to $x = -\frac{C}{D}ct$, which implies

$$\frac{C}{D} = -\frac{v}{c}. \quad (1.4)$$

Define

$$\beta \equiv \frac{v}{c}, \quad (1.5)$$

so that

$$C = -\beta D. \quad (1.6)$$

For the same reason, $x = 0$ must correspond to $x' = -vt'$. On the other hand, it corresponds to $x' = \frac{C}{A}ct' = -\frac{D}{A}vt'$, which implies

$$D = A. \quad (1.7)$$

In other words, the transformation must read

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} A & B \\ -\beta A & A \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \quad (1.8)$$

1.2 Newtonian Geometry of Space and Time

Newton assumed that time is time is time, i.e., that time is *absolute*: $t' = t$. Thus,

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \quad (1.9)$$

This is what is called a *Galilean boost*.

The inverse relationship is simply

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}. \quad (1.10)$$

Let (ct_1, x_1, y_1, z_1) and (ct_2, x_2, y_2, z_2) be the spacetime coordinates of two events as measured in S . As measured in S' , the coordinates just have primes on them. Let $\Delta t = t_2 - t_1$, $\Delta x = x_2 - x_1$, etc. You should show that Galilean boosts leave the following unchanged:

$$\Delta t, \quad \Delta r^2 \equiv \Delta x^2 + \Delta y^2 + \Delta z^2. \quad (1.11)$$

Furthermore, the Galilean boost transformation leads to our standard and intuitive notion of velocity addition. The velocity of an object as measured in S' is related to that in S in the familiar way:

$$u'_x = \frac{dx'}{dt'} = \frac{d}{dt}(x - \beta ct) = u_x - v. \quad (1.12)$$

1.3 The Spacetime Geometry of Special Relativity

Einstein assumed something different: that the speed of light is the same in all inertial reference frames. This assumption did not come out of the blue. We will discuss shortly where it came from. But, for now, let's just run with it and see what happens to the coordinate transformation. Consider a photon traveling in the $+x$ direction. In S , the path of the photon (its worldline) is described by the equation

$$(ct)^2 - x^2 = 0, \quad (1.13)$$

and $y = 0$ and $z = 0$, of course. This can be written in a seemingly silly way as

$$\begin{pmatrix} ct & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} = 0. \quad (1.14)$$

If the speed of this photon is to be c in S' as well, its worldline in S' must be described by the equation

$$0 = (ct')^2 - x'^2 = \begin{pmatrix} ct' & x' \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}. \quad (1.15)$$

If the speed of the photon in S' where $c' \neq c$, then ct' would instead read $c't'$.

On the other hand, using (1.8), the above equation reads

$$\begin{aligned} 0 &= \begin{pmatrix} ct & x \end{pmatrix} \begin{pmatrix} A & -\beta A \\ B & A \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} A & B \\ -\beta A & A \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} \\ &= \begin{pmatrix} ct & x \end{pmatrix} \begin{pmatrix} (1 - \beta^2)A^2 & A(B + \beta A) \\ A(B + \beta A) & B^2 - A^2 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \end{aligned} \quad (1.16)$$

Comparing this with (1.14), we first conclude (from the off-diagonal components of the matrix in the center) that

$$B = -\beta A, \quad (1.17)$$

in which case,

$$(1 - \beta^2)A^2 \begin{pmatrix} ct & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} = 0. \quad (1.18)$$

Since the transformation matrix must become the identity matrix in the limit $\beta \rightarrow 0$, we must set

$$A = \gamma \equiv \frac{1}{\sqrt{1 - \beta^2}}. \quad (1.19)$$

Then, the transformation matrix becomes the *Lorentz boost*:

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \quad (1.20)$$

You should show that you get back the Galilean boost transformation in (1.9) in the $c \rightarrow \infty$ limit. Lorentz transformations no longer keep Δt and Δr^2 , defined in (1.11), invariant. Instead, only the combination

$$\Delta s^2 \equiv (c\Delta t)^2 - \Delta r^2 \quad (1.21)$$

remains invariant. This is called the invariant interval. You should verify this fact.

Let us define the lightcone coordinates

$$x^\pm \equiv \frac{ct \pm x}{\sqrt{2}}, \quad (1.22)$$

the inverse relations of which read

$$ct = \frac{x^+ + x^-}{\sqrt{2}}, \quad x = \frac{x^+ - x^-}{\sqrt{2}}. \quad (1.23)$$

These relations may be written as

$$\begin{pmatrix} x^+ \\ x^- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}, \quad \begin{pmatrix} ct \\ x \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \end{pmatrix}. \quad (1.24)$$

Therefore, in lightcone coordinates, a Lorentz boost transformation reads

$$\begin{aligned} \begin{pmatrix} x^{+'} \\ x^{-'} \end{pmatrix} &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} \\ &= \frac{\gamma}{\sqrt{2}} \begin{pmatrix} 1-\beta & 1-\beta \\ 1+\beta & -(1+\beta) \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \end{aligned} \quad (1.25)$$

Finally,

$$\begin{aligned} \begin{pmatrix} x^{+'} \\ x^{-'} \end{pmatrix} &= \frac{\gamma}{\sqrt{2}} \begin{pmatrix} 1-\beta & 1-\beta \\ 1+\beta & -(1+\beta) \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \end{pmatrix} \\ &= \gamma \begin{pmatrix} 1-\beta & 0 \\ 0 & 1+\beta \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \end{pmatrix}. \end{aligned} \quad (1.26)$$

This is a bit simpler since it is in diagonal form: $x^{\pm'} = \gamma(1 \mp \beta)x^{\pm}$.

1.4 Wherefore, Einstein?

Why did Einstein think to assume that the speed of light is constant in all inertial reference frames. After all, it flies in the face of the Galilean velocity addition formula (1.12) and leads to all manner of weird and wacky consequences (e.g., time dilation, length contraction, loss of simultaneity, etc.) It's not completely crazy and it certainly didn't come out of the blue. It was inspired by Maxwell's equations of electricity and magnetism. I'm not going to write all four of them down. Instead, I will only write down the wave equation, which must be satisfied by the electric and magnetic fields that describe a plane wave of light traveling in vacuum in the $\pm x$ direction. You should remind yourself how you derive this wave equation from Maxwell's equations in vacuum. We have

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) \psi = 0, \quad (1.27)$$

where ψ represents any Cartesian component of the electric or the magnetic field. It is convenient to convert to lightcone coordinates:

$$\begin{aligned} \partial_{\pm} &\equiv \frac{\partial}{\partial x^{\pm}} = \frac{\partial(ct)}{\partial x^{\pm}} \frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial x}{\partial x^{\pm}} \frac{\partial}{\partial x} \\ &= \frac{1}{\sqrt{2}} \left(\frac{1}{c} \frac{\partial}{\partial t} \pm \frac{\partial}{\partial x} \right). \end{aligned} \quad (1.28)$$

Therefore, the wave equation may be written more simply as

$$\partial_+ \partial_- \psi = 0. \quad (1.29)$$

The most general transformation that leaves this equation invariant is

$$\begin{pmatrix} x^{+'} \\ x^{-'} \end{pmatrix} = \begin{pmatrix} C_+ & 0 \\ 0 & C_- \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \end{pmatrix} + \begin{pmatrix} \delta x^+ \\ \delta x^- \end{pmatrix}, \quad (1.30)$$

where $C_{\pm}$ and $\delta x^{\pm}$ are constants such that

$$C_+ C_- = 1. \quad (1.31)$$

The shift represents translation long the lightcone. We must focus just on the homogeneous part of this transformation, with $\delta x^\pm = 0$. Then, the solution to the above equation may be parametrized precisely as in the Lorentz transformation (1.26) written in lightcone coordinates:

$$C_\pm = \gamma(1 \mp \beta), \quad (1.32)$$

which, indeed, satisfies (1.31). But, we already showed that the Lorentz boost keeps the speed of light invariant. Hence, Einstein's assumption. Note that we have also implicitly assumed another one of Einstein's postulates: the fundamental laws of physics are the same in all inertial reference frames.

1.5 The Interval and the Lightcone

Depending on the sign of Δs^2 between two spacetime events, those events are called time-like, light-like, or space-like separated:

$$\begin{aligned} \Delta s^2 > 0, & \quad \text{time-like interval,} \\ \Delta s^2 = 0, & \quad \text{light-like or null interval,} \\ \Delta s^2 < 0, & \quad \text{space-like interval.} \end{aligned} \quad (1.33)$$

The span of all points which are connected to an event via a light-like interval is called the light-cone at the point. The light-cone is split into two: the past ($\Delta t < 0$) and future ($\Delta t > 0$) lightcone. These identifications are invariant under Lorentz transformations. In other words, a Lorentz transformation will never turn a space-like interval into a time-like one and vice-versa.

1.6 Time Dilation

Show Figure 1.4. Then, quickly move on to Figure 1.5, then Figure 1.8. The upper hyperboloid is the surface of “one-tick”: when a clock in each reference frame ticks once. From the perspective of S , all of the other clocks are ticking more slowly than its own and the faster the clock is moving, the slower is it ticking. However, if you draw a line going through B and parallel to the x' axis, then it will intersect the ct' axis *above* the point D . Therefore, from the point of view of S' , the clock in S is ticking more slowly! In other words, the slowing down of clocks effect is reciprocal! This is the statement of *time dilation*.

Let $\Delta\tau$ be the *proper* time interval between two events. Here, *proper* means in the rest frame in which the two events happen at the same point in space. What is the time interval between the two events as measured in some other inertial frame moving at the speed v relative to the rest frame? You can use the invariance of Δs^2 or the Lorentz transformations to work out that it is

$$\Delta t = \gamma \Delta\tau. \quad (1.34)$$

Since $\gamma \geq 1$, the time interval in any other reference frame is longer than in the rest frame. That's time dilation.

1.7 Rear Clocks Run Ahead

PSA: I'm saying rear clocks run *ahead*, not *faster*. Those are different things!

I'll leave this as a problem set exercise. It's easy to see that rear clocks run ahead just from the spacetime diagram. The worldlines of the front and back of the train from the perspective of the platform are parallel lines with slope $1/\beta$ and horizontal separation equal to L/γ (due to length

contraction). That horizontal slice is what the frame, S , of the platform observes as the state of the train at a given moment in time. However, the train in its proper rest frame, S' , is actually not a horizontal slice but a slice parallel to the x' axis, which is a line of slope β : that is, a fixed proper time slice. You can immediately see that the rear clock reads 0 from the perspective of S at an earlier time compared to when the front clock reads 0. The problem set question asks you to work out precisely by how much.

1.8 Understanding Time Dilation

A common stumbling block related to time dilation is the following. In the previous example, from the train frame, Charles zooms by, he passes Bob first, at which point their clocks are both set to 0, and then, a time later, Charles passes Alice. In the train frame, the distance that Charles has to travel is L and his speed is v . Therefore, the time it takes for this to happen is $\frac{L}{v}$ in the train frame. In other words, when Charles passes Alice, Alice's clock reads the time $\frac{L}{v}$.

During that time, from Alice's or Bob's perspective, the amount of time that has elapsed on Charles' clock is $\frac{L}{\gamma v}$ (i.e., less time than $\frac{L}{v}$) because from their perspective, Charles' clock runs more slowly compared to theirs.

However, from Charles' perspective, Alice's and Bob's clocks run more slowly compared to his clock! So, how can it be that Alice's clock registers $\frac{L}{v}$, which is more than what Charles' clock registers, namely $\frac{L}{\gamma v}$, if Alice's clock runs more slowly than Charles' clock (from Charles' perspective)?

I'll leave it to you to ponder this question in the problem set.

1.9 Length Contraction

Length contraction can be derived as a corollary of time dilation, or vice versa, so I won't bother to say any more about it. Just remember the pole-in-the-barn paradox, which combines length contraction and loss of simultaneity.

1.10 The Doppler Effect

So far, we haven't worried about the time it takes for information about the state of one part of a system to travel to a particular observer. Basically, we imagine that the spacetime of a particular reference frame is filled with observers at rest in that frame. They record everything that happens at their point in space and only later do they convene to reconstruct a global picture of what happened throughout space and time.

For the Doppler effect, however, we do have to worry about the travel time of the signal because the frequency or wavelength of that signal is precisely the matter of concern.

Suppose S and S' are conventionally defined (the axes of S and S' are aligned, S' is moving in the x -direction with speed v and their spacetime origins coincide). Let their common origin correspond to the event of a first light pulse sent from S to S' . It is emitted and received immediately. One period T later, from the perspective of S , they send another pulse towards S' . However, in this time, S' has moved a distance vT away. So, the light pulse now has to catch up. Let δt be the time it takes (in S) to catch up:

$$c\Delta t = v(T + \Delta t) \quad \implies \quad \Delta t = \frac{v}{c - v} T. \quad (1.35)$$

Thus, the spacetime coordinates (in S) of the event when the second light pulse catches up with the

observer in S' is

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} c(T + \Delta t) \\ c\Delta t \end{pmatrix} = \begin{pmatrix} \frac{c}{c-v} cT \\ \frac{v}{c-v} cT \end{pmatrix} = \frac{cT}{1-\beta} \begin{pmatrix} 1 \\ \beta \end{pmatrix}. \quad (1.36)$$

The corresponding coordinates in S' are

$$\begin{aligned} \begin{pmatrix} ct' \\ x' \end{pmatrix} &= \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \frac{cT}{1-\beta} \begin{pmatrix} 1 \\ \beta \end{pmatrix} \\ &= \frac{\gamma cT}{1-\beta} \begin{pmatrix} 1-\beta^2 \\ 0 \end{pmatrix} \\ &= \sqrt{\frac{1+\beta}{1-\beta}} cT \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \end{aligned} \quad (1.37)$$

Of course, it had better be the case that $x' = 0$ because both the reception of the first signal and the reception of the second signal happen at the location of the observer in S' , which, by definition, is not moving in S' . So, both events must have the same x' coordinates. Since we set the origin to be the reception of the first signal, this x' coordinate is 0. Hence, the period between the two pulses as measured in S' is

$$T' = \sqrt{\frac{1+\beta}{1-\beta}} T \quad \implies \quad \frac{T'}{T} = \sqrt{\frac{1+\beta}{1-\beta}}. \quad (1.38)$$

Frequency, on the other hand, is inversely proportional to the period. Therefore, the relationship between the frequency of a signal observed by an observer moving away from an emitter (or the other way round) and the emitted frequency is

$$\frac{\nu'}{\nu} = \sqrt{\frac{1-\beta}{1+\beta}}. \quad (1.39)$$

1.11 Lorentz Transformations as Four-Dimensional ‘Rotations’

It is useful to define the so-called *rapidity* parameter of a Lorentz boost:

$$\psi = \tanh^{-1}(\beta). \quad (1.40)$$

You should easily be able to show that the Lorentz transformation can be written as

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh \psi & -\sinh \psi \\ -\sinh \psi & \cosh \psi \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \quad (1.41)$$

This looks a bit similar to an actual rotation matrix. For example, if we were to rotate in the y - z plane by an angle θ in the counter-clockwise direction (from the positive y -axis towards the positive z -axis), then

$$\begin{pmatrix} y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix}. \quad (1.42)$$

Check for yourself that, if we define (the subscript E stands for ‘Euclidean’)

$$t_E \equiv it, \quad (1.43)$$

and similarly for the primed time, and we define the imaginary angle

$$\phi \equiv i\psi, \quad (1.44)$$

then

$$\begin{pmatrix} ct'_E \\ x' \end{pmatrix} = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} ct_E \\ x \end{pmatrix}. \quad (1.45)$$

Now, this really does look like a rotation, but nevermind that the Euclidean time and the angle ϕ are both technically imaginary!

This is nice because we can clearly see that the composition of two boosts in the same direction gives a boost whose rapidity is simply the sum of the separate rapidities of the two boosts for the same reason that the composition of two rotations in the same direction is a rotation by an angle equal to the sum of the angles of the two rotations.

1.12 Velocity Addition

From the previous statement, you should be able to derive the first equation in eq. (1.13) of the textbook, which relates the velocity of something as measured in S' to its velocity as measured in S . Then, you need only remember that the transformation of the velocity components transverse to the velocity of S' with respect to S just picks up a γ_v in the denominator.

1.13 Acceleration

The derivation of the transformation of the acceleration under a Lorentz boost, given in sec. 1.13 of the textbook, is a little bit tedious, but nevertheless straightforward. You can see that if the acceleration vanishes in one inertial reference frame, then it vanishes in all inertial reference frames.

1.14 Worldline

The path in spacetime of a particle is called its *worldline*. For comparison, the path of string in spacetime is called its *worldsheet*, and, in general, for higher dimensions, this is called the *worldvolume*. Imagine a reference frame that moves along with the particle so that the particle is always instantaneously at rest in that frame. This is called the IRF or *Instantaneous Rest Frame* of the particle. In the IRF, the infinitesimal invariant interval along the particle's worldline is

$$ds^2 = (cd\tau)^2, \quad (1.46)$$

where τ is the proper time in the IRF. In any other coordinate system associated with a reference frame S ,

$$(cd\tau)^2 = ds^2 = (cdt)^2 - dx^2 - dy^2 - dz^2 = (cdt)^2(1 - \beta^2), \quad (1.47)$$

where

$$(\beta c)^2 = v^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2. \quad (1.48)$$

In other words

$$d\tau = dt/\gamma, \quad (1.49)$$

where it is understood that $v = v(t)$ is a function of time, and, thus, so is γ .

The total proper time elapsed between spacetime points A and B is

$$\Delta\tau = \int_A^B d\tau = \int_A^B \frac{dt}{\gamma(t)} = \int_A^B dt \sqrt{1 - \frac{v(t)^2}{c^2}}. \quad (1.50)$$

1.15 Action Principle

In moving from point A to point B , we would like particles to follow Fermat's principle of minimum time: they should follow the path that minimizes the total proper time. Therefore, it would make sense to set the *action* of the particle to be proportional to the proper time elapsed, or the integral of the invariant interval. Of course, the action should have units of energy $\times$ time, so that it becomes dimensionless once divided by Planck's constant $\hbar$, which also has units of energy $\times$ time, or angular momentum. As it is, $\int ds$ has units of length. To get units of energy $\times$ time, we must multiply by a mass and a speed. There is no relevant mass besides the mass of the particle, and there is no other fundamental intrinsic speed besides the speed of light. Thus,

$$S \sim \pm mc \int ds. \quad (1.51)$$

Since we want the particle to *minimize* the action, we pick the minus sign. Otherwise, *any* path in the future lightcone would have a larger proper time than the path of a lightray and so everything would follow the path of a lightray, which is absurd. Thus,

$$S = -mc \int ds. \quad (1.52)$$

Let us write this as

$$S = -mc \int \sqrt{(cdt)^2 - dr^2} = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}} \approx \int dt \left(\frac{1}{2}mv^2 - mc^2 \right). \quad (1.53)$$

We recognize the kinetic energy $T = \frac{1}{2}mv^2$ part of the Lagrangian and the rest-mass energy part of the potential energy $V = mc^2$.

Chapter 2

Electromagnetism and Gravity “Derived”

We will see that Maxwell’s theory of electromagnetism and Einstein’s theory of general relativity can both be “derived” by making the point particle described in the previous section interact with the external world in a coordinate-invariant way.

The factor mc^2 is the rest-mass energy, it doesn’t count as a potential energy describing the interaction of the particle with anything external to itself. In the non-relativistic case, adding such a potential energy is straightforward:

$$\int dt \left(\frac{1}{2}mv^2 - mc^2 \right) \rightarrow \int dt \left(\frac{1}{2}mv^2 - mc^2 - V(x) \right). \quad (2.1)$$

Can we come up with a modification of the relativistic point-particle action (1.52), such that its non-relativistic limit reproduces (2.1)? Yes, in fact there are two approaches:

$$\text{Approach 1: } S = -mc \int \sqrt{(cdt)^2 - dr^2} - \int dt V(x), \quad (2.2a)$$

$$\text{Approach 2: } S = -mc \int \sqrt{\left(1 + \frac{2V(x)}{mc^2}\right)(cdt)^2 - dr^2}. \quad (2.2b)$$

Convince yourself that the leading non-relativistic limits of both of the above approaches is (2.1).

Before we tackle each of these approaches in turn, let us introduce some notation. We denote the spacetime coordinates by

$$x^\mu, \quad \mu = 0, 1, 2, 3, \quad (2.3)$$

or up to however many spatial coordinates there are, where

$$x^0 = ct, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z, \quad (2.4)$$

and so on. Next, define the Minkowski metric

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (2.5)$$

Then, the square of the invariant interval may be written as

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu, \quad (2.6)$$

where the Einstein summation rule is assumed to be in effect: a pair of repeated indices, one lower and one upper, are summed over. In other words

$$\eta_{\mu\nu} dx^\mu dx^\nu = \sum_{\mu,\nu=0}^3 \eta_{\mu\nu} dx^\mu dx^\nu = (cdt)^2 - dr^2. \quad (2.7)$$

2.1 Approach 1: Electromagnetism

Write (2.2a) as

$$S = -mc \int ds - \int dt V(x). \quad (2.8)$$

There is a problem! The first term, involving the invariant interval, is invariant under Lorentz transformations. The second term, however, is not: time and space mix under a Lorentz boost. We have to relativistically complete dt to dx^μ and $V(x)$ to a vector field A_μ that may depend on time and space:

$$S \rightarrow -mc \int ds - \int dx^\mu A_\mu(x), \quad (2.9)$$

By fiat, since under a coordinate transformation $x^\mu \rightarrow x'^\mu(x)$, the vector dx^μ transforms as

$$dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu, \quad (2.10)$$

it must be the case that $A_\mu(x)$ transforms as

$$A'_\mu(x') = \frac{\partial x^\nu}{\partial x'^\mu} A_\nu(x), \quad (2.11)$$

such that

$$\int dx'^\mu A'_\mu(x') = \int dx^\nu \frac{\partial x'^\mu}{\partial x^\nu} \frac{\partial x^\rho}{\partial x'^\mu} A_\rho(x) = \int dx^\nu \delta_\nu^\rho A_\rho(x) = \int dx^\nu A_\nu(x) \quad (2.12)$$

is, indeed, invariant.

This smells of electromagnetism, where A_μ is the 4-vector potential with respect to which the electric and magnetic fields are derived. We will find that this is, indeed, the case.

Parametrize the action in terms of the proper time of the particle in its IRF:

$$S = -mc \int d\tau \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} - \int d\tau A_\mu(x(\tau)) \frac{dx^\mu}{d\tau}. \quad (2.13)$$

Now, vary this with respect to x :

$$\begin{aligned} \delta S &= -mc \int d\tau \frac{\eta_{\mu\nu} \frac{d\delta x^\mu}{d\tau} \frac{dx^\nu}{d\tau}}{\sqrt{\eta_{\rho\lambda} \frac{dx^\rho}{d\tau} \frac{dx^\lambda}{d\tau}}} - \int d\tau \left(A_\mu \frac{d\delta x^\mu}{d\tau} + \frac{\partial A_\nu}{\partial x^\mu} \delta x^\mu \frac{dx^\nu}{d\tau} \right) \\ &= \int d\tau \left\{ mc \eta_{\mu\nu} \frac{d}{d\tau} \left(\frac{\frac{dx^\nu}{d\tau}}{\sqrt{\eta_{\rho\lambda} \frac{dx^\rho}{d\tau} \frac{dx^\lambda}{d\tau}}} \right) + \frac{dA_\mu}{d\tau} - (\partial_\mu A_\nu) \frac{dx^\nu}{d\tau} \right\} \delta x^\mu \\ &= \int d\tau \left\{ mc \eta_{\mu\nu} \frac{d}{d\tau} \left(\frac{\frac{dx^\nu}{d\tau}}{\sqrt{\eta_{\rho\lambda} \frac{dx^\rho}{d\tau} \frac{dx^\lambda}{d\tau}}} \right) - (\partial_\mu A_\nu - \partial_\nu A_\mu) \frac{dx^\nu}{d\tau} \right\} \delta x^\mu. \end{aligned} \quad (2.14)$$

The equation of motion is given by setting the content of the $\{\dots\}$ equal to 0. We can write this as

$$\frac{mc}{\sqrt{\eta_{\rho\lambda} \frac{dx^\rho}{d\tau} \frac{dx^\lambda}{d\tau}}} \eta_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} - \frac{mc}{(\eta_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau})^{3/2}} \eta_{\mu\nu} \eta_{\rho\lambda} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} \frac{d^2 x^\lambda}{d\tau^2} - F_{\mu\nu} \frac{dx^\nu}{d\tau} = 0, \quad (2.15)$$

where the electromagnetic field strength is defined to be

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.16)$$

Raise the μ index:

$$\frac{mc}{\sqrt{\eta_{\rho\lambda} \frac{dx^\rho}{d\tau} \frac{dx^\lambda}{d\tau}}} \frac{d^2 x^\mu}{d\tau^2} - \frac{mc}{(\eta_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau})^{3/2}} \eta_{\rho\lambda} \frac{dx^\mu}{d\tau} \frac{dx^\rho}{d\tau} \frac{d^2 x^\lambda}{d\tau^2} - F^\mu{}_\nu \frac{dx^\nu}{d\tau} = 0, \quad (2.17)$$

In the non-relativistic limit, $\tau \approx t$ and we can simply approximate

$$\sqrt{\eta_{\rho\lambda} \frac{dx^\rho}{d\tau} \frac{dx^\lambda}{d\tau}} \approx c. \quad (2.18)$$

The middle term in (2.17) is higher-order in v/c , and so we just have

$$m \frac{d^2 x^\mu}{d\tau^2} = F^\mu{}_\nu \frac{dx^\nu}{d\tau}. \quad (2.19)$$

This is nothing but Newton's second law $F = ma$, where the force is the Lorentz force due to an electromagnetic field. Let the momentum be

$$p^\mu = m \frac{dx^\mu}{d\tau} = \gamma m(c, \mathbf{v}) = \left(\frac{E}{c}, \mathbf{p} \right). \quad (2.20)$$

Then,

$$\frac{dp^\mu}{d\tau} = F^\mu{}_\nu \frac{dx^\nu}{d\tau} \implies \frac{dp^\mu}{dt} = F^\mu{}_\nu \frac{dx^\nu}{dt}. \quad (2.21)$$

Let us break this up into the zeroth component:

$$\frac{1}{c} \frac{dE}{dt} = F^0{}_i v^i. \quad (2.22)$$

This had better be equivalent to the equation

$$\frac{dE}{dt} = \mathbf{E} \cdot \mathbf{v}, \quad (2.23)$$

which is, indeed, the work done by an electric field. I'll leave it to you to show that this is the case.

The spatial components, on the other hand, read

$$\frac{dp^i}{dt} = F^i{}_0 c + F^i{}_j \frac{dx^j}{dt}. \quad (2.24)$$

Again, you should convince yourself that this is the same as the Lorentz force equation:

$$\frac{d\mathbf{p}}{dt} = \mathbf{E} + \mathbf{v} \times \mathbf{B}. \quad (2.25)$$

These are the equations of motion for a point charge of mass m and unit charge in the presence of an electric and magnetic field. For a charge q , we should simply multiply the RHS above equations by q . Clearly, that simply means that (2.13) is just modified to

$$S = -mc \int d\tau \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} - q \int d\tau A_\mu(x(\tau)) \frac{dx^\mu}{d\tau}. \quad (2.26)$$

2.2 Approach 2: Gravity

Now, let's consider the action (2.2b). Note that all we've done is promote $\eta_{\mu\nu}$ to a possibly spacetime-dependent metric $g_{\mu\nu}(x)$ given by

$$g_{\mu\nu} = \left(1 + \frac{2V}{mc^2}\right) \delta_\mu^0 \delta_\nu^0 - \delta_\mu^i \delta_\nu^j \delta_{ij}. \quad (2.27)$$

In other words, we have made the change

$$S = -mc \int \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} \longrightarrow S = -mc \int \sqrt{g_{\mu\nu}(x) dx^\mu dx^\nu}. \quad (2.28)$$

Now, this describes the motion of a massive point particle in a possibly curved background spacetime with metric $g_{\mu\nu}(x)$!

Chapter 3

Manifolds

A manifold M of dimension n is a topological space such that every point $m \in M$ is contained in at least one element of the open cover $\{U_\alpha\}$, where $\alpha \in I$ is some label that indexes the open sets comprising the open cover. Additionally, each open set U_α must be accompanied with a continuous bijective map $\phi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$, such that the image of U_α under the map ϕ_α is an open subset of $\mathbb{R}^n$. If U_α and U_β intersect, then ϕ_α and ϕ_β map the overlap region into $\mathbb{R}^n$ in two different ways. The map $\phi_\alpha \circ \phi_\beta^{-1}$ (called a transition function) must be homeomorphism from $\phi_\beta(U_\alpha \cap U_\beta)$ to $\phi_\alpha(U_\alpha \cap U_\beta)$. If all these maps are smooth, then the manifold is called a smooth manifold. That's usually what we will assume.

Let's give a simple example: the circle S^1 is a smooth one-dimensional manifold. There are many ways to define the circle *as a topological space*. For this purpose, any loop that does not intersect itself is a circle. So, we can define a circle to be the interval $[0, 2\pi]$ with 0 and 2π identified as the same point. Even easier would be to define S^1 as $\mathbb{R}/\sim$ where $x \sim y$ if $y - x = 2\pi n$ for some integer n . You can pretty much see how to use $\sim$ to map open sets in S^1 to open sets in $\mathbb{R}$. For example, let the open cover consist of

$$U_1 = (-\frac{\epsilon}{2}, \frac{2\pi}{3} + \frac{\epsilon}{2})/\sim, \quad (3.1a)$$

$$U_2 = (\frac{2\pi}{3} - \frac{\epsilon}{2}, \frac{4\pi}{3} + \frac{\epsilon}{2})/\sim, \quad (3.1b)$$

$$U_3 = (\frac{4\pi}{3} - \frac{\epsilon}{2}, 2\pi + \frac{\epsilon}{2})/\sim. \quad (3.1c)$$

Then, the map $\phi_1 : U_1 \rightarrow \mathbb{R}$, and ϕ_2 and ϕ_3 are all just "remove the $\sim$." Let us check the transition function for $U_1 \cap U_2 = (\frac{2\pi}{3} - \frac{\epsilon}{2}, \frac{2\pi}{3} + \frac{\epsilon}{2})/\sim$. This is map both by ϕ_1 and ϕ_2 into $(\frac{2\pi}{3} - \frac{\epsilon}{2}, \frac{2\pi}{3} + \frac{\epsilon}{2})$, so clearly, $\phi_1 \circ \phi_2^{-1}$ is simply the identity map from $\phi_2(U_1 \cap U_2)$ to $\phi_1(U_1 \cap U_2)$. The same holds for the intersection $U_2 \cap U_3$. Now, for $U_1 \cap U_3$, its image under ϕ_1 is $(-\frac{\epsilon}{2}, \frac{\epsilon}{2})$, whereas its image under ϕ_3 is $(2\pi - \frac{\epsilon}{2}, 2\pi + \frac{\epsilon}{2})$. The map $\phi_1 \circ \phi_3^{-1}$ is simply given by

$$\phi_1 \circ \phi_3^{-1} : (2\pi - \frac{\epsilon}{2}, 2\pi + \frac{\epsilon}{2}) \rightarrow (-\frac{\epsilon}{2}, \frac{\epsilon}{2}), \quad \phi_1 \circ \phi_3^{-1}(x) = x - 2\pi. \quad (3.2)$$

It is obvious that all of these maps are smooth. Thus, S^1 is a smooth manifold.

3.1 Vectors

Consider all possible one-dimensional curves that go through a particular point $p \in M$. The span of all possible tangent vectors of those curves at the point p is the tangent space at p , $T_p(M)$. You can easily see that this forms a vector space since these tangent vectors may be multiplied by a real number and may be added with each other and the zero vector is the point p itself. Any smooth curve in M is parametrized by an affine parameter λ , which we can think of as a map $\gamma : \mathbb{R} \rightarrow M$ from λ to

$\gamma(\lambda)$. The tangent to a curve is, of course generated by the derivative $\frac{d}{d\lambda}$. Suppose we have a map $f : M \rightarrow \mathbb{R}$ and I would like to know what the directional derivative of f is along the curve γ . This is often denoted $\frac{df}{d\lambda}$, except that this notation does not really make any sense: the input of f is a point in the manifold M , not a value of λ . Of course, what we really mean is

$$\text{“abuse of notation”} : \frac{df}{d\lambda}, \quad \text{“correct”} : \frac{d}{d\lambda}(f \circ \gamma), \quad (3.3)$$

where $f \circ \gamma$ is a map from $\mathbb{R}$ to $\mathbb{R}$.

To compute this, we take a detour into $\mathbb{R}^n$, which give the local coordinates on the manifold:

$$\frac{d}{d\lambda}(f \circ \gamma) = \frac{d}{d\lambda}[(f \circ \phi^{-1}) \circ (\phi \circ \gamma)]. \quad (3.4)$$

Now, $\phi \circ \gamma$ is a map $\mathbb{R} \rightarrow \mathbb{R}^n$ and we know perfectly well how to define the derivative of that with respect to λ . Meanwhile, $f \circ \phi^{-1}$ is a map from $\mathbb{R}^n \rightarrow \mathbb{R}$ and we also know how to define the derivative of that:

$$\frac{d}{d\lambda}(f \circ \gamma) = \frac{d(\phi \circ \gamma)^\mu}{d\lambda} \frac{\partial(f \circ \phi^{-1})}{\partial x^\mu} = \frac{dx^\mu(\lambda)}{d\lambda} \partial_\mu(f \circ \phi^{-1}). \quad (3.5)$$

Again, by abuse of notation, this is sometimes written as

$$\frac{df}{d\lambda} = \frac{dx^\mu}{d\lambda} \partial_\mu f. \quad (3.6)$$

On the LHS, it is taken for granted that f should really be $f \circ \gamma$ because the $\frac{d}{d\lambda}$ indicates that we must evaluate f not on the whole of M , but only on the curve γ . On the RHS, by forgetting the ϕ^{-1} , we are essentially conflating the point on the manifold with the coordinates used to describe it. In (3.6), it is understood that x^μ in $\frac{dx^\mu}{d\lambda}$ is the image of λ under the map $\phi \circ \gamma$, that is, it is the image of the curve in M under ϕ , it is not the entire coordinate system itself. On the other hand, the x^μ in $\partial_\mu = \frac{\partial}{\partial x^\mu}$ is the entire coordinate system.

It follows from (3.6) that the generator $\frac{d}{d\lambda}$ of tangent vectors can be written as a superposition of the ∂_μ 's. In other words, ∂_μ forms a basis for tangent vectors, and thus for the tangent space at a point on the manifold. This is what is called a *coordinate basis* for $T_p(M)$. It is not the only basis we could have chosen, but it is fairly simple and intuitive. It also makes the transformation under coordinate transformations particularly simple:

$$x'^\mu(x^\nu) \implies \partial'_\mu = \frac{\partial x^\nu}{\partial x'^\mu} \partial_\nu, \quad (3.7)$$

where $\partial x / \partial x'$ is the inverse of $\partial x' / \partial x$.

Thus, any vector field V can be expanded in terms of the coordinate basis $\{\partial_\mu\}$:

$$V = V^\mu \partial_\mu, \quad (3.8)$$

and it is understood to act on any function f on the manifold (with the understanding that we really mean $f \circ \phi^{-1}$). That is,

$$V(f) \equiv V^\mu \partial_\mu f. \quad (3.9)$$

The *commutator* or *Lie bracket* of two vectors is another vector that acts on functions in the following way:

$$[X, Y](f) \equiv X(Y(f)) - Y(X(f)). \quad (3.10)$$

In components, $[X, Y] = [X, Y]^\mu \partial_\mu$, where

$$[X, Y]^\mu = X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu. \quad (3.11)$$

3.2 One-forms and Tensors

When you have a vector space, like $T_p(M)$, you can always define the dual vector space $T_p^*(M)$ to be the vector space of all linear maps $\omega : T_p(M) \rightarrow \mathbb{R}$. A linear map means

$$\omega(aV + bW) = a\omega(V) + b\omega(W), \quad (3.12)$$

where $V, W \in T_p(M)$ and $a, b \in \mathbb{R}$. You can easily show that the space of all such linear maps itself forms a real vector space, where scalar multiplication and addition are defined rather obviously. Dual vectors are also called *one-forms*.

Since we have a basis for vectors, namely ∂_μ , we can define a basis of one-forms by how they act on the basis for vectors. The *coordinate basis* for one-forms is just the one that acts orthonormally on ∂_μ : define dx^μ to act as

$$dx^\mu(\partial_\nu) = \frac{\partial x^\mu}{\partial x^\nu} = \delta_\nu^\mu. \quad (3.13)$$

The action of dx^μ is extended to all of $T_p(M)$ by linearity and $T_p^*(M)$ is defined as the span of the basis $\{dx^\mu\}$. These transform in the inverse way to ∂_μ :

$$dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu. \quad (3.14)$$

One can also think of T_p as the dual of T_p^* . In other words, a vector can be thought of as a linear map on the space of one-forms. In general, you can consider a multilinear map from k copies of T_p^* and ℓ copies of T_p to $\mathbb{R}$. That is called a tensor of rank (k, ℓ) :

$$T : \underbrace{T_p^* \times \cdots \times T_p^*}_{k \text{ times}} \times \underbrace{T_p \times \cdots \times T_p}_{\ell \text{ times}} \rightarrow \mathbb{R}. \quad (3.15)$$

This map just needs to be linear in each one of its factors separately. The tensor product of a (k, ℓ) tensor with a (k', ℓ') tensor is a $(k + k', \ell + \ell')$ tensor:

$$\begin{aligned} (T \otimes S)(\omega^{(1)}, \dots, \omega^{(k+k')}, V^{(1)}, \dots, V^{(\ell+\ell')}) \\ = T(\omega^{(1)}, \dots, \omega^{(k)}, V^{(1)}, \dots, V^{(\ell)}) S(\omega^{(k+1)}, \dots, \omega^{(k+k')}, V^{(\ell+1)}, \dots, V^{(\ell+\ell')}). \end{aligned} \quad (3.16)$$

The tensor product is the obvious and natural extension to the space of tensor of ordinary scalar multiplication in $\mathbb{R}$. You can obviously extend this to finitely many tensor products and we can use tensor products of basis vectors and one-forms to write the components of a tensor:

$$T = T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_\ell} \partial_{\mu_1} \otimes \cdots \otimes \partial_{\mu_k} \otimes dx^{\nu_1} \otimes \cdots \otimes dx^{\nu_\ell}. \quad (3.17)$$

Each upper index transforms like a one-form (3.14) and each lower index transforms like a vector (3.7). Each pair (one upper and one lower) of indices that are contracted with each other (i.e., have the same index label and are thus summed over by the Einstein summation convention) together cancel each other's transformation and form a scalar component, thereby reducing the rank. For example, if $T^\mu{}_\nu$ is a $(1, 1)$ tensor, then $T^\mu{}_\mu$ is a scalar.

We've already seen a $(0, 2)$ tensor, namely the metric $g_{\mu\nu}$, and a $(2, 0)$ tensor, namely the inverse metric $g^{\mu\nu}$. We've seen another example of a $(0, 2)$ -tensor, namely the electromagnetic field strength $F_{\mu\nu}$. This one is actually also what is called a 2-form because it is antisymmetric.

3.3 Tensor Density

Another important quantity is the *Levi-Civita symbol*, which, for n -dimensional spacetime is given by:

$$\tilde{\epsilon}_{\mu_1 \dots \mu_n} = \begin{cases} +1, & \mu_1 \dots \mu_n \text{ is an even permutation of } 0, 1, \dots, n-1, \\ -1, & \mu_1 \dots \mu_n \text{ is an odd permutation of } 0, 1, \dots, n-1, \\ 0, & \text{otherwise.} \end{cases} \quad (3.18)$$

Except, this is not actually a tensor. This is a cautionary tale to point out that not everything with indices is a tensor: it must transform like a tensor in order to actually be a tensor. To work out how the Levi-Civita symbol transforms, note that it enters into the definition of the determinant of a matrix:

$$\tilde{\epsilon}_{\mu_1 \dots \mu_n} \det M = \tilde{\epsilon}_{\nu_1 \dots \nu_n} M^{\nu_1}_{\mu_1} \dots M^{\nu_n}_{\mu_n}. \quad (3.19)$$

Therefore, if M is the matrix $M^\nu_\mu = \frac{\partial x^\nu}{\partial x'^\mu}$ for some coordinate transformation $x'(x)$, then

$$\tilde{\epsilon}'_{\mu_1 \dots \mu_n} = \det \left(\frac{\partial x'^\rho}{\partial x^\sigma} \right) \tilde{\epsilon}_{\nu_1 \dots \nu_n} \frac{\partial x^{\nu_1}}{\partial x'^{\mu_1}} \dots \frac{\partial x^{\nu_n}}{\partial x'^{\mu_n}}. \quad (3.20)$$

This is *almost* the transformation rule for a $(0, n)$ tensor were it not for the Jacobian factor out front. Because of that Jacobian factor, the Levi-Civita symbol is called a tensor *density*. The exponent of the Jacobian is called the *weight* of the tensor density. That is, the Levi-Civita symbol is a $(0, 4)$ tensor density of weight 1.

Another example of a tensor density is the determinant of the metric. Since the metric transforms like a $(0, 2)$ tensor,

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} g_{\rho\sigma}(x), \quad (3.21)$$

it follows that the determinant of the metric transforms as

$$g'(x') \equiv \det[g'_{\mu\nu}(x')] = \det \left(\frac{\partial x^\rho}{\partial x'^\mu} \right) \det \left(\frac{\partial x^\sigma}{\partial x'^\nu} \right) \det[g_{\rho\sigma}(x)] = \det \left(\frac{\partial x^\rho}{\partial x'^\mu} \right) \det \left(\frac{\partial x^\sigma}{\partial x'^\nu} \right) g(x). \quad (3.22)$$

The determinants here are inverses of the Jacobian factor in (3.20). Therefore, the weight in this case is -2 . That is, the determinant of the metric is a scalar density of weight -2 . Note that there is nothing particularly special about the metric for this purpose. The determinant of any $(0, 2)$ tensor is a scalar density of weight -2 . Likewise, the determinant of any $(2, 0)$ tensor is a scalar density of weight 2. Of course, for a 2-form, like $F_{\mu\nu}$, the determinant obviously vanishes anyway, so the its determinant is indeed a scalar density of weight -2 , but a very trivial one.

Since the determinant of the metric is a scalar density of weight -2 , its square root is a scalar density of weight -1 . Therefore, if $\tilde{T}$ is a tensor density of weight w , then

$$T = |g|^{w/2} \tilde{T} \quad (3.23)$$

is a tensor. In particular, we can define the Levi-Civita *tensor*:

$$\epsilon_{\mu_1 \dots \mu_n} \equiv \sqrt{|g|} \tilde{\epsilon}_{\mu_1 \dots \mu_n}. \quad (3.24)$$

The inverse Levi-Civita tensor is

$$\epsilon^{\mu_1 \dots \mu_n} = \frac{1}{\sqrt{|g|}} \tilde{\epsilon}^{\mu_1 \dots \mu_n}. \quad (3.25)$$

It is important to be able to contract subsets of the indices of the Levi-Civita tensor and its inverse:

$$\epsilon^{\mu_1 \dots \mu_p \alpha_1 \dots \alpha_{n-p}} \epsilon_{\mu_1 \dots \mu_p \beta_1 \dots \beta_{n-p}} = (-1)^s p!(n-p)! \delta_{\beta_1}^{[\alpha_1} \dots \delta_{\beta_{n-p}}^{\alpha_{n-p}]}, \quad (3.26)$$

where s is the number of negative eigenvalues of the metric and the square bracket enclosing a number of indices is called “antisymmetrization.” This process is defined as follows:

$$T^{[\mu_1 \cdots \mu_m]} \equiv \frac{1}{m!} \sum_{\sigma \in S_m} (-1)^{\text{sgn}(\sigma)} T^{\mu_{\sigma(1)} \cdots \mu_{\sigma(m)}}, \quad (3.27)$$

where S_m is the set of permutations of $1, \dots, m$ and $\text{sgn}(\sigma)$ is the sign of the permutation $\sigma \in S_m$, which is $+1$ if the permutation is an even permutation and -1 if it is an odd permutation. Let’s show some simple examples:

$$T^{[\mu\nu]} = \frac{T^{\mu\nu} - T^{\nu\mu}}{2}, \quad (3.28)$$

and

$$T^{[\mu\nu\rho]} = \frac{1}{6} (T^{\mu\nu\rho} + T^{\nu\rho\mu} + T^{\rho\mu\nu} - T^{\nu\mu\rho} - T^{\mu\rho\nu} - T^{\rho\nu\mu}). \quad (3.29)$$

In contrast, if we use round brackets, then we mean symmetrization:

$$T^{(\mu_1 \cdots \mu_m)} \equiv \frac{1}{m!} \sum_{\sigma \in S_m} T^{\mu_{\sigma(1)} \cdots \mu_{\sigma(m)}}, \quad (3.30)$$

with examples being the same as above but with all plus signs.

If you need to exclude some indices from the symmetrization or antisymmetrization, you can indicate this by putting vertical bars around the excluded indices:

$$T^{(\mu|\rho|\nu)} = \frac{T^{\mu\rho\nu} + T^{\nu\rho\mu}}{2}. \quad (3.31)$$

3.4 Differential Forms, Wedge Product, and Cohomology

We have already seen an example, $F_{\mu\nu}$, of a $(0, 2)$ tensor which is also a 2-form because it happens to be antisymmetric. More generally, a $(0, p)$ tensor which is fully antisymmetric in the p indices is a p -form. Scalars and $(0, 1)$ tensors are automatically 0- and 1-forms, respectively. The space of all p -forms is denoted Λ^p and all p -form *fields* on a manifold M is denoted $\Lambda^p(M)$. The dimension of $\Lambda^p(M)$ is $\binom{n}{p} = \frac{n!}{p!(n-p)!}$, where n is the dimension of the manifold. We can denote the space of all forms Λ , with Λ naturally being graded by the degree of the form: $\Lambda = \bigoplus_{p=0}^{\infty} \Lambda_p$.

We can define a product on the space of all forms called the *wedge product*. The wedge product of a p -form A and a q -form B is a $(p + q)$ -form defined to be

$$[A \wedge B]_{\mu_1 \cdots \mu_{p+q}} = \frac{(p+q)!}{p!q!} A_{[\mu_1 \cdots \mu_p} B_{\mu_{p+1} \cdots \mu_{p+q}]}. \quad (3.32)$$

For the basis one-forms, set $A = A^{(\mu)} = A_{\nu}^{(\mu)} dx^{\nu}$ with

$$A_{\nu}^{(\mu)} = \delta_{\nu}^{\mu}, \quad (3.33)$$

so that $A^{(\mu)} = dx^{\mu}$, and similarly for B . Then,

$$[A^{(\mu)} \wedge A^{(\nu)}]_{\rho\sigma} = (dx^{\mu} \wedge dx^{\nu})_{\rho\sigma} = \frac{2!}{1!1!} \delta_{[\rho}^{\mu} \delta_{\sigma]}^{\nu} = \delta_{\rho}^{\mu} \delta_{\sigma}^{\nu} - \delta_{\sigma}^{\mu} \delta_{\rho}^{\nu}, \quad (3.34)$$

or

$$dx^{\mu} \wedge dx^{\nu} = dx^{\mu} \otimes dx^{\nu} - dx^{\nu} \otimes dx^{\mu} = 2dx^{[\mu} \otimes dx^{\nu]}. \quad (3.35)$$

In general,

$$A \wedge B = (-1)^{pq} B \wedge A. \quad (3.36)$$

We can define another map called the differential $d : \Lambda_p \rightarrow \Lambda_{p+1}$ given by

$$(dA)_{\mu_1 \dots \mu_{p+1}} = (p+1) \partial_{[\mu_1} A_{\mu_2 \dots \mu_{p+1}]} \quad (3.37)$$

For a scalar, it simply gives the gradient:

$$d\phi = (\partial_\mu \phi) dx^\mu. \quad (3.38)$$

The differential operator distributes over the wedge product of a p -form A and a q -form B as follows:

$$d(A \wedge B) = (dA) \wedge B + (-1)^p A \wedge (dB). \quad (3.39)$$

You can show quite easily that

$$d^2 = 0. \quad (3.40)$$

A p -form A is *closed* if $dA = 0$ and is *exact* if $A = dB$ for some $(p-1)$ -form B . Closed p -form fields constitute a vector space $Z^p(M)$, since a linear combination of closed forms is also closed. The same is true for exact forms, called $B^p(M)$. Therefore, we can define quotient vector space, which is also a group under addition, of course, called the *de Rham cohomology group*,

$$H^p(M) = \frac{Z^p(M)}{B^p(M)}. \quad (3.41)$$

Interestingly, this cohomology gives you measures of the topology of the manifold. For example, flat space is topologically equivalent to $\mathbb{R}^4$ whose homologies for $p > 0$ all vanish. That is, locally, all closed forms are exact (except for scalars, of course). The cohomology groups of S^2 , the 2-sphere, for example, are $H^0(S^2) = H^2(S^2) = \mathbb{R}$ and $H^1(S^2) = 0$. Obviously, the cohomology groups only go up to the dimension of the manifold itself. In case you couldn't already tell that the sphere is topologically distinct from flat space, this is one way to definitively prove this fact: their cohomology groups are different, so they are topologically distinct.

The cohomology groups of T^2 , the 2-torus, are $H^0(T^2) = H^2(T^2) = \mathbb{R}^2$ and $H^1(T^2) = \mathbb{R}^2$. Therefore, T^2 is topologically distinct from the sphere and flat space, which I think is also pretty obvious, but always nice to prove rigorously.

3.5 Hodge Duality

There exists a different map, called the Hodge star operator, that maps p -forms to $(n-p)$ -forms:

$$(\star A)_{\mu_1 \dots \mu_{n-p}} = \frac{1}{p!} \epsilon^{\nu_1 \dots \nu_p}{}_{\mu_1 \dots \mu_{n-p}} A_{\nu_1 \dots \nu_p}. \quad (3.42)$$

Using the contraction rules for the Levi-Civita tensor, you can show that

$$\star^2 = (-1)^{s+p(n-p)}, \quad (3.43)$$

where, again, s is the number of negative eigenvalues of the metric.

In 3D Euclidean space, there is no essential difference between a one-form and a vector (as there are no sign flips when we raise or lower indices). The wedge product of two vectors can thus be thought of as a 2-form and the Hodge dual of that would be a one-form, or a vector. Except, it's not really a vector, not in the stricter sense, which is to say that it should flip sign under a reflection over a plane

perpendicular to it. Because of the epsilon tensor in the definition of the Hodge star, the Hodge dual of the wedge product of two vectors is actually a pseudo-vector, a vector in every way except that it does *not* switch sign under a reflection over a plane perpendicular to it. In fact, this construction is nothing but the usual *cross product* of two vectors. In fact, in any other dimension higher than 3, the cross product of two vectors can be defined, but it gives a 2-form, not a vector. It just so happens that, in three dimensions, that 2-form is Hodge dual to a pseudo-vector.

3.6 Electricity and Magnetism

We can see a number of the objects and maps that we have defined play a role in the theory of electricity and magnetism. As already mentioned, the field strength is a 2-form:

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu. \quad (3.44)$$

The relationship between F and the gauge field $A = A_\mu dx^\mu$ is simply an exterior derivative:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \Longleftrightarrow \quad F = dA. \quad (3.45)$$

In fact, the defining quality of F is that it be closed:

$$dF = 0 \quad \Longleftrightarrow \quad \partial_{[\mu} F_{\nu\rho]} = 0. \quad (3.46)$$

This is nothing but the Bianchi identity and is equivalent to Gauss' law for magnetism and Faraday's law:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (3.47)$$

The fact that we can write F as d of a one-form A is actually only true for manifolds with $H^2(M) = 0$, which happens to be the case for flat space. But this is not true for S^2 or T^2 , so there are, in fact, obstructions to writing E & M on those manifolds purely in terms of a vector gauge field.

Meanwhile, the sourced equations (Gauss' law and Ampère's law) can be written as

$$d \star F = 4\pi \star J \quad (3.48)$$

where $J = J_\mu dx^\mu$ is the source, $J_0 = \rho$ and J_i is the vector current. This one equation is equivalent to

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad -\dot{\mathbf{E}} + \nabla \times \mathbf{B} = 4\pi\mathbf{J}. \quad (3.49)$$

Furthermore, you can also check that the usual Lagrangian term

$$F_{\mu\nu} F^{\mu\nu} \quad \Longleftrightarrow \quad F \wedge \star F, \quad (3.50)$$

and, in fact, there is another term you could write down, which is $F \wedge F$. This is called the “theta” term because it is often introduced with a coefficient denoted θ . Such a term induces CP violation. For E&M, with chiral fermions, it turns out that you can rotate this θ term away. It becomes important in QCD, however.

Let us verify these results. First, let us compute $F_{\mu\nu}$ from scratch starting from

$$A = A_\mu dx^\mu = \phi dt - \mathbf{A} \cdot d\mathbf{x}, \quad (3.51)$$

where ϕ is the usual scalar potential and $\mathbf{A}$ is the vector potential. Note that

$$A^\mu = (\phi, \mathbf{A}), \quad A_\mu = (\phi, -\mathbf{A}). \quad (3.52)$$

Then,

$$(dA)_{\mu\nu} = (1+1)\partial_{[\mu}A_{\nu]} = F_{\mu\nu} \implies F = dA. \quad (3.53)$$

On the other hand,

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = -\partial_i \phi - (\dot{\mathbf{A}})_i = E_i, \quad (3.54a)$$

$$F_{12} = \partial_1 A_2 - \partial_2 A_1 = -\partial_1(\mathbf{A})_2 + \partial_2(\mathbf{A})_1 = -B_3, \quad (3.54b)$$

$$F_{13} = \partial_1 A_3 - \partial_3 A_1 = -\partial_1(\mathbf{A})_3 + \partial_3(\mathbf{A})_1 = B_2, \quad (3.54c)$$

$$F_{23} = \partial_2 A_3 - \partial_3 A_2 = -\partial_2(\mathbf{A})_3 + \partial_3(\mathbf{A})_2 = -B_1. \quad (3.54d)$$

Thus,

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix}. \quad (3.55)$$

Let us compute the Hodge dual of F .

$$(\star F)_{01} = \epsilon^{23}_{01} F_{23} = \epsilon_{2301} F_{23} = \epsilon_{0123} F_{23}. \quad (3.56)$$

We have to be careful in the definition of the Levi-Civita tensor. We have make a conventional choice to agree with standard notation. That choice is

$$\epsilon^{0123} = 1. \quad (3.57)$$

Everything else follows from this. For example,

$$\epsilon_{0123} = -\epsilon^{0123} = -1. \quad (3.58)$$

Thus,

$$(\star F)_{01} = -F_{23} = B_1. \quad (3.59)$$

For the rest,

$$(\star F)_{02} = \epsilon^{13}_{02} F_{13} = \epsilon_{1302} F_{13} = -\epsilon_{0123} F_{13} = F_{13} = B_2, \quad (3.60a)$$

$$(\star F)_{03} = \epsilon^{21}_{03} F_{21} = \epsilon_{2103} F_{21} = -\epsilon_{0123} F_{21} = F_{21} = B_3, \quad (3.60b)$$

$$(\star F)_{12} = \epsilon^{03}_{12} F_{03} = -\epsilon_{0312} F_{03} = -\epsilon_{0123} F_{03} = F_{03} = E_3, \quad (3.60c)$$

$$(\star F)_{13} = \epsilon^{02}_{13} F_{02} = -\epsilon_{0213} F_{02} = \epsilon_{0123} F_{02} = -F_{02} = -E_2, \quad (3.60d)$$

$$(\star F)_{23} = \epsilon^{01}_{23} F_{01} = -\epsilon_{0123} F_{01} = F_{01} = E_1, \quad (3.60e)$$

or

$$(\star F)_{\mu\nu} = \begin{pmatrix} 0 & B_1 & B_2 & B_3 \\ -B_1 & 0 & E_3 & -E_2 \\ -B_2 & -E_3 & 0 & E_1 \\ -B_3 & E_2 & -E_1 & 0 \end{pmatrix}. \quad (3.61)$$

Let us compute $F \wedge \star F$:

$$(F \wedge \star F)_{\mu\nu\rho\sigma} = \frac{(2+2)!}{2!2!} F_{[\mu\nu} (\star F)_{\rho\sigma]} = 6 F_{[\mu\nu} (\star F)_{\rho\sigma]}. \quad (3.62)$$

Since this is a 4-form, we can just as well plug in $\mu\nu\rho\sigma = 0123$ and all other orders of the indices can be retrieved from this one:

$$\begin{aligned}
(F \wedge \star F)_{0123} &= 6F_{[01}(\star F)_{23]} \\
&= F_{01}(\star F)_{23} - F_{02}(\star F)_{13} - F_{03}(\star F)_{21} - F_{21}(\star F)_{03} - F_{31}(\star F)_{20} + F_{23}(\star F)_{01} \\
&= E_1(E_1) - E_2(-E_2) - E_3(-E_3) - B_3(B_3) - (-B_2)(-B_2) + (-B_1)(B_1) \\
&= |\mathbf{E}|^2 - |\mathbf{B}|^2.
\end{aligned} \tag{3.63}$$

That's precisely 2 times the electromagnetic Lagrangian density. Therefore, we can write

$$\mathcal{L}_{\text{EM}} d\text{vol} = \frac{1}{2} F \wedge \star F, \tag{3.64}$$

where the volume element is

$$d\text{vol} = dx^0 \wedge dx^1 \wedge dx^2 \wedge dx^3. \tag{3.65}$$

Meanwhile,

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix}, \tag{3.66}$$

so

$$F_{\mu\nu} F^{\mu\nu} = 2(F_{0i} F^{0i} + F_{ij} F^{ij}) = 2[E_i(-E_i) + |\mathbf{B}|^2] = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2). \tag{3.67}$$

Hence,

$$F \wedge \star F = -\frac{1}{2} (F_{\mu\nu} F^{\mu\nu}) d\text{vol}, \tag{3.68}$$

and

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \tag{3.69}$$

Meanwhile,

$$\begin{aligned}
(F \wedge F)_{0123} &= 6F_{[01}F_{23]} \\
&= 2(F_{01}F_{23} + F_{02}F_{31} + F_{03}F_{12}) \\
&= 2[E_1(-B_1) + E_2(-B_2) + E_3(-B_3)] \\
&= -2\mathbf{E} \cdot \mathbf{B}.
\end{aligned} \tag{3.70}$$

On the other hand, note that

$$(\star F)^{\mu\nu} = \begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & E_3 & -E_2 \\ B_2 & -E_3 & 0 & E_1 \\ B_3 & E_2 & -E_1 & 0 \end{pmatrix}, \tag{3.71}$$

and so

$$F_{\mu\nu} (\star F)^{\mu\nu} = 2[F_{0i} (\star F)^{0i} + F_{ij} (\star F)^{ij}] = -4\mathbf{E} \cdot \mathbf{B}. \tag{3.72}$$

Therefore,

$$F \wedge F = \frac{1}{2} F_{\mu\nu} (\star F)^{\mu\nu} d\text{vol}. \tag{3.73}$$

It is clear that, while $F \wedge \star F$ preserves spatial parity, $F \wedge F$ does not: under parity, $\mathbf{A} \rightarrow -\mathbf{A}$ and $\nabla \rightarrow -\nabla$, but $\mathbf{B} \rightarrow \mathbf{B}$ because $\mathbf{B} = \nabla \times \mathbf{A}$. On the other hand, $\mathbf{E} \rightarrow -\mathbf{E}$ and so $F \wedge F \rightarrow -F \wedge F$.

Let us vary $\mathcal{L}_{\text{EM}}$ with respect to A_μ :

$$\begin{aligned} \frac{\delta \mathcal{L}_{\text{EM}}}{\delta A_\mu} &= -\frac{1}{2} F^{\rho\sigma} \left(\partial_\rho \frac{\delta A_\sigma}{\delta A_\mu} - \partial_\sigma \frac{\delta A_\rho}{\delta A_\mu} \right) \\ &= \frac{1}{2} \partial_\rho F^{\rho\mu} - \frac{1}{2} \partial_\sigma F^{\mu\sigma} \\ &= \partial_\nu F^{\nu\mu}, \end{aligned} \quad (3.74)$$

where we integrated by parts to get to the middle line above. Let us write out the components of this:

$$\partial_\mu F^{\mu 0} = \partial_i (E_i) = \nabla \cdot \mathbf{E}, \quad (3.75a)$$

$$\partial_\mu F^{\mu 1} = \partial_0 F^{01} + \partial_2 F^{21} + \partial_3 F^{31} = -\dot{E}_1 + \partial_2 B_3 + \partial_3 (-B_2) = (-\dot{\mathbf{E}} + \nabla \times \mathbf{B})_1, \quad (3.75b)$$

$$\partial_\mu F^{\mu 2} = \partial_0 F^{02} + \partial_1 F^{12} + \partial_3 F^{32} = -\dot{E}_2 + \partial_1 (-B_3) + \partial_3 B_1 = (-\dot{\mathbf{E}} + \nabla \times \mathbf{B})_2, \quad (3.75c)$$

$$\partial_\mu F^{\mu 3} = \partial_0 F^{03} + \partial_1 F^{13} + \partial_2 F^{23} = -\dot{E}_3 + \partial_1 B_2 + \partial_2 (-B_1) = (-\dot{\mathbf{E}} + \nabla \times \mathbf{B})_3. \quad (3.75d)$$

In vacuum, the above all vanish, which just give Gauss' law and Ampère's law. To add sources, we just add a term to the Lagrangian:

$$\mathcal{L}_{\text{source}} = -4\pi A_\mu J^\mu, \quad (3.76)$$

where J^μ is built out of the charge density and current density,

$$J^\mu = (\rho, \mathbf{J}). \quad (3.77)$$

Thus, the Euler-Lagrange equations of motion become

$$\partial_\nu F^{\nu\mu} = 4\pi J^\mu, \quad (3.78)$$

or, in components:

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad \nabla \times \mathbf{B} - \dot{\mathbf{E}} = 4\pi\mathbf{J}, \quad (3.79)$$

which are Gauss' law and Ampère's law in the presence of sources.

On the other hand, from (3.64), we have

$$\begin{aligned} \delta \mathcal{L}_{\text{EM}} d\text{vol} &= \frac{1}{2} \delta (F \wedge \star F) \\ &= \frac{1}{2} (d\delta A) \wedge \star F + \frac{1}{2} F \wedge \star d\delta A \end{aligned} \quad (3.80)$$

The first term is easy to integrate by parts and becomes $-\frac{1}{2} \delta A \wedge d \star F$. The second term, however, has a $\star$ in the way. However, when ω is a p -form and η is an $(n-p)$ -form, then $\omega \wedge \eta$ is an n -form, proportional to the volume element. In this case, the result is the same as $(-1)^{p(n-p)} (\star \eta) \wedge (\star \omega)$. Thus,

$$F \wedge \star d\delta A = (d\delta A) \wedge \star F, \quad (3.81)$$

and we just lose the factor of 2:

$$\delta \mathcal{L}_{\text{EM}} d\text{vol} = (d\delta A) \wedge \star F = d(\delta A \wedge \star F) + \delta A \wedge d \star F \stackrel{\text{IBP}}{=} \delta A \wedge d \star F. \quad (3.82)$$

Next, with J^μ defined in (3.77), we have

$$J_\mu = (\rho, -\mathbf{J}), \quad (3.83)$$

and we can consider the 1-form $J = J_\mu dx^\mu$. Since we want a source Lagrangian that is linear in A and J , and both are 1-forms, we have Hodge dual one of them, say J :

$$(\star J)_{\mu\nu\rho} = \epsilon^\sigma_{\mu\nu\rho} J_\sigma, \quad (3.84)$$

and

$$(A \wedge \star J)_{0123} = \frac{4!}{1!3!} A_{[0} (\star J)_{123]} = 4A_{[0} \epsilon^\mu_{123]} J_\mu. \quad (3.85)$$

We expand this into the nonzero terms:

$$\begin{aligned} (A \wedge \star J)_{0123} &= A_0 \epsilon^0_{123} J_0 - A_1 \epsilon^1_{023} J_1 - A_2 \epsilon^2_{103} J_2 - A_3 \epsilon^3_{120} J_3 \\ &= \epsilon_{0123} A_0 J_0 + \epsilon_{1023} A_1 J_1 + \epsilon_{2103} A_2 J_2 + \epsilon_{3120} A_3 J_3 \\ &= -A_0 J_0 - \epsilon_{0123} A_1 J_1 - \epsilon_{0123} A_2 J_2 - \epsilon_{0123} A_3 J_3 \\ &= -A_0 J_0 + A_1 J_1 + A_2 J_2 + A_3 J_3 \\ &= -(A_0 J^0 + A_1 J^1 + A_2 J^2 + A_3 J^3) \\ &= -A_\mu J^\mu. \end{aligned} \quad (3.86)$$

Hence,

$$\mathcal{L}_{\text{source}} d\text{vol} = 4\pi A \wedge \star J, \quad (3.87)$$

and

$$\delta \mathcal{L}_{\text{source}} d\text{vol} = 4\pi (\delta A) \wedge \star J, \quad (3.88)$$

Together with (3.82), this leads to the equation of motion

$$d \star F = -4\pi \star J. \quad (3.89)$$

Taking the star of this gives

$$\star d \star F = -4\pi J, \quad (3.90)$$

where we note that $\star^2 = 1$ on a 1-form given a metric with 3 negative eigenvalues.

Let us check the 0-component:

$$4\pi J_\mu = -\frac{1}{3!} \epsilon^{\nu\rho\sigma}_\mu (2+1) \partial_{[\nu} \frac{1}{2!} \epsilon^{\alpha\beta}_{\rho\sigma]} F_{\alpha\beta} = -\frac{1}{4} \epsilon^{\nu\rho\sigma}_\mu \epsilon^{\alpha\beta}_{\rho\sigma} \partial_\nu F_{\alpha\beta}, \quad (3.91)$$

where we removed the antisymmetrization symbol on the lower indices ν , ρ , and σ because they are contracted with the upper versions of themselves, which are contained in an ϵ tensor, and are therefore already antisymmetric. Let us do some raising and lowering and shifting around of indices:

$$4\pi J^\mu = -\frac{1}{4} \epsilon^{\rho\sigma\nu\mu} \epsilon_{\rho\sigma\alpha\beta} \partial_\nu F^{\alpha\beta}. \quad (3.92)$$

Using (3.26) with $s = 3$, $p = 2$ and $n - p = 2$, we get

$$4\pi J^\mu = -\frac{1}{4} (-1) 2! 2! \delta_\alpha^{[\nu} \delta_\beta^{\mu]} \partial_\nu F^{\alpha\beta} = \partial_\nu F^{\nu\mu}, \quad (3.93)$$

where we again removed the antisymmetrization on ν and μ because that is equivalent to antisymmetrization in α and β , which is automatic since $F^{\alpha\beta}$ is already antisymmetric. Indeed, this is consistent with (3.78).

3.7 Integration

Integration over a subregion Σ of dimension p in the manifold M of dimension n can be thought of as a linear map from the space of p -form fields, $\Lambda^p(M)$ to the real numbers, $\mathbb{R}$:

$$\int_{\Sigma} : \Lambda^p(M) \rightarrow \mathbb{R}. \quad (3.94)$$

The integration measure is nothing but the Levi-Civita tensor!:

$$\begin{aligned} \epsilon &= \epsilon_{\mu_1 \dots \mu_n} dx^{\mu_1} \otimes \dots \otimes dx^{\mu_n} \\ &= \frac{1}{n!} \epsilon_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \\ &= \frac{1}{n!} \sqrt{|g|} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \\ &= \sqrt{|g|} dx^0 \wedge \dots \wedge dx^{n-1} \\ &\equiv \sqrt{|g|} d^n x. \end{aligned} \quad (3.95)$$

So, the integral of a function $\phi(x)$ is simply

$$I = \int_{\Sigma} \epsilon \phi(x). \quad (3.96)$$

Actually, the way to interpret this is to say that we are integrating the n -form $\phi(x) \epsilon$. In general, a p -form can naturally be integrated over a p -dimensional submanifold of M .

Let's look again at Maxwell theory and particularly at the equation of motion (3.89). We can turn this into an integral expression rather than a differential one simply by integrating it, but over what space are we supposed to integrate? Well, J is a 1-form, so $\star J$ is a 3-form, which means that this must be integrated over a 3-dimensional submanifold, M_3 , of spacetime:

$$\int_{M_3} d \star F = -4\pi \int_{M_3} \star J. \quad (3.97)$$

The divergence theorem allows us to change the LHS into a surface integral over the boundary $M_2 = \partial M_3$:

$$\int_{M_2} \star F = -4\pi \int_{M_3} \star J. \quad (3.98)$$

When M_3 is a purely spatial volume, this is nothing but the integral form of Gauss' law.

Let's give a simple example. Let J be the purely a charge density (no current density) given by

$$\rho = \lambda \delta(x) \delta(y) = \lambda \frac{\delta(r)}{2\pi r}, \quad (3.99)$$

where $r = \sqrt{x^2 + y^2}$ is the cylindrical radial distance to the z -axis. Then,

$$(\star J)_{\mu\nu\rho} = \epsilon^0_{\mu\nu\rho} J_0 = \epsilon_{0\mu\nu\rho} \rho, \quad (3.100)$$

so

$$(\star J)_{r\phi z} = \epsilon_{0r\phi z} \rho = -\epsilon^{0r\phi z} \rho = -\sqrt{-g} \rho = -\rho r, \quad (3.101)$$

or

$$\star J = -\rho r dr \wedge d\phi \wedge dz. \quad (3.102)$$

Let M_2 be a cylinder centered on the z -axis, of radius r and length ℓ . Then, M_3 is the volume of the interior of this cylinder and

$$\int_{M_3} \star J = - \int_0^r \rho(r') r' dr' \int_0^{2\pi} d\phi \int_{z_0}^{z_0+\ell} dz = -\lambda \ell. \quad (3.103)$$

Meanwhile,

$$\begin{aligned} \star F &= E_1 dx^2 \wedge dx^3 + E_2 dx^3 \wedge dx^1 + E_3 dx^1 \wedge dx^2 \\ &= E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy \\ &= (E_r \cos \phi - E_\phi \sin \phi)(\sin \phi dr + r \cos \phi d\phi) \wedge dz \\ &\quad + (E_r \sin \phi + E_\phi \cos \phi) dz \wedge (\cos \phi dr - r \sin \phi d\phi) \\ &\quad + E_z (\cos \phi dr - r \sin \phi d\phi) \wedge (\sin \phi dr + r \cos \phi d\phi) \\ &= r(E_r d\phi \wedge dz + E_z dr \wedge d\phi) + E_\phi dz \wedge dr. \end{aligned} \quad (3.104)$$

Cylindrical symmetry tells us that $E_\phi = E_z = 0$ and

$$E_r = E_r(r). \quad (3.105)$$

Therefore,

$$\int_{M_2} \star F = r E_r \int_0^{2\pi} d\phi \int_{z_0}^{z_0+\ell} dz = 2\pi r E_r \ell, \quad (3.106)$$

and (3.98) implies

$$\boxed{E_r = \frac{2\lambda}{r}}. \quad (3.107)$$

Indeed, that is the electric field due to an infinitely long straight wire of linear charge density λ .

Let us consider an example where $M_3 = [t_0, t_0 + T] \times \Sigma$, where $[t_0, t_0 + T]$ is a time interval of length T and Σ is a closed 2-dimensional spatial surface. Then, (3.98) becomes the generalized Biot-Savart law (generalized to include a possibly time-dependent electric field).

Consider the example of a wire with zero net charge carrying a current I in the $+z$ direction. So,

$$\begin{aligned} \star F &= B_x dt \wedge dx + B_y dt \wedge dy + B_z dt \wedge dz \\ &= (B_r \cos \phi - B_\phi \sin \phi) dt \wedge (\cos \phi dr - r \sin \phi d\phi) \\ &\quad + (B_r \sin \phi + B_\phi \cos \phi) dt \wedge (\sin \phi dr + r \cos \phi d\phi) + B_z dt \wedge dz \\ &= B_r dt \wedge dr + r B_\phi dt \wedge d\phi + B_z dt \wedge dz. \end{aligned} \quad (3.108)$$

The only problem is that $dt \wedge d\phi$ (think of it as a cross-product) points *into* the interior of M_3 . This is the opposite of the convention (i.e., $\partial M_3 = M_2$ is thought of as pointing *outwards* from the interior of M_3). Therefore, we write

$$\star F = -B_r dr \wedge dt - r B_\phi d\phi \wedge dt - B_z dz \wedge dt. \quad (3.109)$$

The symmetry of the current-carrying wire implies $B_r = B_z = 0$ and

$$B_\phi = B_\phi(r), \quad (3.110)$$

and so integrating over M_2 when Σ is a circle of radius r centered on the z -axis gives

$$\int_{M_2} \star F = -r B_\phi \int_0^{2\pi} d\phi \int_{t_0}^{t_0+T} dt = -2\pi r B_\phi T. \quad (3.111)$$

Meanwhile, all components of the current density vanish except for

$$J_z = -(\mathbf{J})_z = -I \frac{\delta(r)}{2\pi r}, \quad (3.112)$$

and thus,

$$(\star J)_{0r\phi} = \epsilon^z{}_{0r\phi} J_z = \epsilon^{z0r\phi} J_z = -\epsilon^{0r\phi z} J_z = -\sqrt{-g} J_z = -r J_z = \frac{I}{2\pi} \delta(r). \quad (3.113)$$

Then,

$$\int_{M_3} \star J = \frac{I}{2\pi} \int_{t_0}^{t_0+T} dt \int_0^r \delta(r') dr' \int_0^{2\pi} d\phi = IT, \quad (3.114)$$

and so,

$$B_\phi = \frac{2I}{r}. \quad (3.115)$$

Let's go back to the previous example, in which case

$$E_x = \frac{2\lambda}{r} \cos \phi = \frac{2\lambda x}{r^2}, \quad E_y = \frac{2\lambda}{r} \sin \phi = \frac{2\lambda y}{r^2}, \quad E_z = 0. \quad (3.116)$$

Therefore,

$$F_{\mu\nu} = \frac{2\lambda}{r^2} \begin{pmatrix} 0 & x & y & 0 \\ -x & 0 & 0 & 0 \\ -y & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3.117)$$

Suppose we make a coordinate transformation to a boosted frame where the boost is by a speed v in the $-z$ -direction:

$$t' = \gamma(t + \beta z), \quad z' = \gamma(z + \beta t), \quad x' = x, \quad y' = y, \quad (3.118)$$

with inverse

$$t = \gamma(t' - \beta z'), \quad z = \gamma(z' - \beta t'), \quad x = x', \quad y = y'. \quad (3.119)$$

Thus,

$$\begin{aligned} F'_{\mu\nu}(x') &= \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} F_{\rho\sigma}(x) \\ &= \frac{\partial t}{\partial x'^\mu} \frac{\partial x}{\partial x'^\nu} F_{01} + \frac{\partial x}{\partial x'^\mu} \frac{\partial t}{\partial x'^\nu} F_{10} + \frac{\partial t}{\partial x'^\mu} \frac{\partial y}{\partial x'^\nu} F_{02} + \frac{\partial y}{\partial x'^\mu} \frac{\partial t}{\partial x'^\nu} F_{20} \\ &= \frac{4\lambda}{r^2} \frac{\partial t}{\partial x'^{[\mu}} (\delta_{\nu]}^1 x + \delta_{\nu]}^2 y) \\ &= \frac{4\lambda}{r'^2} \frac{\partial t}{\partial x'^{[\mu}} (\delta_{\nu]}^1 x' + \delta_{\nu]}^2 y'). \end{aligned} \quad (3.120)$$

I'm particularly interested in a magnetic field component (i.e., spatial-spatial):

$$B'_x = F'_{32} = \frac{2\lambda}{r'^2} \frac{\partial t}{\partial z'} y' = -\frac{2\gamma\lambda\beta y'}{r'^2}, \quad B'_y = F'_{13} = -\frac{2\lambda}{r'^2} \frac{\partial t}{\partial z'} x' = \frac{2\gamma\lambda\beta x'}{r'^2}. \quad (3.121)$$

This magnetic field is pointing in the angular direction (i.e., wrapping around the the z -axis):

$$B'_\phi = -B'_x \sin \phi + B'_y \cos \phi = \frac{2\gamma\lambda\beta}{r'}. \quad (3.122)$$

Now, in this boosted frame, lengths along the z -direction are contracted, whereas charge does not change. Therefore, linear charge density goes up by a factor of γ . In other words, we recognize $\gamma\lambda$ as λ' , the linear charge density in the boosted frame. So, we write

$$B'_\phi = \frac{2\lambda'\beta}{r'}. \quad (3.123)$$

Now, what is $\lambda'\beta$? That is precisely the current, I' , as measured in this boosted frame! Since, we boosted in the $-z$ -direction, the current is in the $+z$ -direction (provided $\lambda > 0$, of course). In other words,

$$B'_\phi = \frac{2I'}{r'}. \quad (3.124)$$

This is precisely the same as (3.115), where we just think of the frame in which there is a current in the wire as boosted with respect to the frame in which there is *no* current in the wire.

Chapter 4

Curvature

For TLDR, let's collect here some of the important formulas:

$$\text{Christoffel symbols:} \quad \Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}), \quad (4.1a)$$

$$\text{Riemann tensor:} \quad R^\rho_{\mu\lambda\nu} = \partial_\lambda \Gamma^\rho_{\mu\nu} - \partial_\nu \Gamma^\rho_{\lambda\mu} + \Gamma^\rho_{\lambda\kappa} \Gamma^\kappa_{\mu\nu} - \Gamma^\rho_{\mu\kappa} \Gamma^\kappa_{\lambda\nu}, \quad (4.1b)$$

$$\text{Ricci tensor:} \quad R_{\mu\nu} = R^\rho_{\mu\rho\nu} = \partial_\rho \Gamma^\rho_{\mu\nu} - \partial_\nu \Gamma^\rho_{\rho\mu} + \Gamma^\rho_{\rho\lambda} \Gamma^\lambda_{\mu\nu} - \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\rho\nu}, \quad (4.1c)$$

$$\text{Ricci scalar:} \quad R = g^{\mu\nu} R_{\mu\nu}, \quad (4.1d)$$

$$\text{Geodesic equation:} \quad \frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\nu} \frac{dx^\rho}{d\lambda} \frac{dx^\nu}{d\lambda} = 0. \quad (4.1e)$$

4.1 Connection

Consider a path $x^\mu(\lambda)$ in the manifold. In principle, we would like to be able to compute derivatives of this path like velocities and accelerations (both appear in the geodesic equation). But, how can we do that? The tangent vector at different points along the path belong to completely different spaces, namely the tangent spaces at those different points. You need what's called a connection to be able to compare a vector in one tangent space to a vector the tangent space at a nearby point. A connection must therefore take one vector, that is a $(1,0)$ -tensor (one upper index), it must take another vector (with one upper index), which serves as the displacement vector from the original point to the nearby point, and then it must spit out a vector (one upper index), namely a vector in the tangent space at the new point. Therefore, the connection should have one upper index (for the output) and two lower indices (to be contracted with the upper indices of the inputs, the vector at the original point and the displacement vector from the original to the new point). Therefore, the connection should look like

$$\Gamma^\rho_{\mu\nu}. \quad (4.2)$$

You can think of the connection as a way to define a derivative that does not depend on your specific choice of coordinates. Why is there a problem in the first place? Consider a vector V^μ , then take its partial derivative. How does this quantity change under a change of coordinates $x \rightarrow x'$? Well,

$$\partial'_\mu V'^\rho = \frac{\partial x^\nu}{\partial x'^\mu} \partial_\nu \left(\frac{\partial x'^\rho}{\partial x^\sigma} V^\sigma \right) = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\sigma} \partial_\nu V^\sigma + \frac{\partial x^\nu}{\partial x'^\mu} \left(\frac{\partial}{\partial x^\nu} \frac{\partial x'^\rho}{\partial x^\sigma} \right) V^\sigma. \quad (4.3)$$

Because of the second term on the RHS above, the partial derivative of a vector is NOT a tensor! It transforms in a way which itself depends on your choice of coordinates! This is obviously problematic.

We need some concept of a “covariant derivative”, traditionally called ∇_μ such that $\nabla_\mu V^\nu$ does transform like a $(1,1)$ -tensor. There is no unique answer to this problem. We have to impose some additional conditions that we would like our solution to satisfy. The first two are pretty obvious:

1. **Linearity:** $\nabla(T + S) = \nabla T + \nabla S$.

2. **Leibniz rule:** $\nabla(T \otimes S) = (\nabla T) \otimes S + T \otimes (\nabla S)$.

Linearity strongly limits the possible form of the covariant derivative. Indeed, it must be linear in the tensor that it acts. For example,

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho_{\mu\nu} V^\nu. \quad (4.4)$$

Let us transform this to a new set of coordinates $x \rightarrow x'$:

$$\nabla'_\mu V'^\rho = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\sigma} \partial_\alpha V^\sigma + \frac{\partial x^\alpha}{\partial x'^\mu} \left(\frac{\partial}{\partial x^\alpha} \frac{\partial x'^\rho}{\partial x^\sigma} \right) V^\sigma + \Gamma'^\rho_{\mu\alpha} \frac{\partial x'^\alpha}{\partial x^\sigma} V^\sigma. \quad (4.5)$$

On the other hand, for this to be a tensor, we must have

$$\nabla'_\mu V'^\rho = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\sigma} \partial_\alpha V^\sigma + \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\sigma} \Gamma^\sigma_{\alpha\lambda} V^\lambda. \quad (4.6)$$

In the last term above, let us switch the two indices σ and λ (after all, they're just dummy variables). Then, we set (4.5) equal to (4.6). The first terms in (4.5) and (4.6) are already equal to each other and drop out. The remaining terms are all directly proportional to V^σ , so we can just drop V^σ altogether. The remaining terms read

$$\Gamma'^\rho_{\mu\alpha} \frac{\partial x'^\alpha}{\partial x^\sigma} + \frac{\partial x^\alpha}{\partial x'^\mu} \left(\frac{\partial}{\partial x^\alpha} \frac{\partial x'^\rho}{\partial x^\sigma} \right) = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\lambda} \Gamma^\lambda_{\alpha\sigma}. \quad (4.7)$$

Multiply this with $\frac{\partial x^\sigma}{\partial x'^\nu}$ and note that $\frac{\partial x'^\alpha}{\partial x^\sigma} \frac{\partial x^\sigma}{\partial x'^\nu} = \delta^\alpha_\nu$ to write

$$\Gamma'^\rho_{\mu\nu} = \frac{\partial x'^\rho}{\partial x^\lambda} \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} \Gamma^\lambda_{\alpha\sigma} - \frac{\partial x^\alpha}{\partial x'^\mu} \left(\frac{\partial}{\partial x^\alpha} \frac{\partial x'^\rho}{\partial x^\sigma} \right) \frac{\partial x^\sigma}{\partial x'^\nu}. \quad (4.8)$$

The first term on the RHS above would be the transformation of $\Gamma^\rho_{\mu\nu}$ if it were a tensor. Due to the second term above, $\Gamma^\rho_{\mu\nu}$ is NOT a tensor! This is why this is called the Christoffel *symbol*, not *tensor*.

Similarly, for a 1-form, we should have

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu + \tilde{\Gamma}^\rho_{\mu\nu} \omega_\rho, \quad (4.9)$$

where $\tilde{\Gamma}^\rho_{\mu\nu}$ is, in principle, some other symbol. We can construct a scalar out of a tensor and a 1-form, namely $\omega_\nu V^\nu$. By the Leibniz rule,

$$\begin{aligned} \nabla_\mu (\omega_\nu V^\nu) &= (\nabla_\mu \omega_\nu) V^\nu + \omega_\nu (\nabla_\mu V^\nu) \\ &= (\partial_\mu \omega_\nu) V^\nu + \tilde{\Gamma}^\rho_{\mu\nu} \omega_\rho V^\nu + \omega_\nu (\partial_\mu V^\nu) + \Gamma^\nu_{\mu\rho} \omega_\nu V^\rho \\ &= \partial_\mu (\omega_\nu V^\nu) + (\tilde{\Gamma}^\rho_{\mu\nu} + \Gamma^\rho_{\mu\nu}) \omega_\rho V^\nu. \end{aligned} \quad (4.10)$$

This is as far as we can go with the assumptions that we used so far. We have to make some more assumptions. A pretty obvious assumption would be that the Kronecker delta is covariantly constant:

3. **Covariant constancy of the identity:** $\nabla_\mu \delta^\rho_\nu = 0$.

This assumption is equivalent to the statement that the covariant constant commutes with contraction. After all, it is the Kronecker delta which is used to contract one lower and one upper index:

3'. **∇ commutes with contractions:** $\nabla_\mu T^\nu_{\nu\rho} = (\nabla T)_\mu{}^\nu_{\nu\rho}$.

Then, because a scalar ϕ does not transform, $\partial_\mu \phi$ is already a tensor and so we propose that the covariant derivative of the scalar is the same as the partial derivative of the scalar:

4. **$\nabla = \partial$ on scalars:** $\nabla_\mu \phi = \partial_\mu \phi$.

Since $\omega_\nu V^\nu$ is a scalar, assumption 4 implies that

$$\nabla_\mu (\omega_\nu V^\nu) = \partial_\mu (\omega_\nu V^\nu), \quad (4.11)$$

and together with (4.10), we conclude that

$$\tilde{\Gamma}^\rho_{\mu\nu} = -\Gamma^\rho_{\mu\nu}. \quad (4.12)$$

All of these rules so far give us the rule for the covariant derivative of any tensor:

$$\begin{aligned} \nabla_\rho T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_\ell} &= \partial_\rho T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_\ell} \\ &+ \Gamma^{\mu_1}_{\rho\lambda} T^{\lambda\mu_2 \dots \mu_k}_{\nu_1 \dots \nu_\ell} + \dots + \Gamma^{\mu_k}_{\rho\lambda} T^{\mu_1 \dots \mu_{k-1}\lambda}_{\nu_1 \dots \nu_\ell} \\ &- \Gamma^\lambda_{\rho\nu_1} T^{\mu_1 \dots \mu_k}_{\lambda\nu_2 \dots \nu_\ell} - \dots - \Gamma^\lambda_{\rho\nu_\ell} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_{\ell-1}\lambda}. \end{aligned} \quad (4.13)$$

These four assumptions are still not enough to completely specify the connection. Suppose we have two connections that satisfy all of these requirements: $\Gamma^\rho_{\mu\nu}$ and $\hat{\Gamma}^\rho_{\mu\nu}$. The first thing we should not is that, while each of these does not satisfy as a tensor, the non-tensorial part of the transformation rule (4.8) cancels when we subtract two connections. In other words,

$$S^\rho_{\mu\nu} \equiv \Gamma^\rho_{\mu\nu} - \hat{\Gamma}^\rho_{\mu\nu} \quad (4.14)$$

does transform as a tensor.

Note that if $\Gamma^\rho_{\mu\nu}$ serves as a connection that satisfies all the above requirements, then so does $\Gamma^\rho_{\nu\mu}$. Since the difference between two connections is a tensor, we can define the so-called torsion tensor

$$T^\rho_{\mu\nu} \equiv \Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\mu} = 2\Gamma^\rho_{[\mu\nu]}. \quad (4.15)$$

We can now impose a fifth requirement:

5. **Torsion-free:** $\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{(\mu\nu)}$.

You can check the transformation rule for the Christoffel symbol (4.8) that if a connection is torsion-free in some coordinate, then it is torsion-free in all coordinates. That is, torsion-freeness is a coordinate-independent property of the connection.

The final condition that we will impose is

6. **Metric compatibility:** $\nabla_\rho g_{\mu\nu} = 0$.

This means that covariant derivatives commute with the raising and lowering of indices.

Finally, with all six requirements, we get a unique solution: the Levi-Civita connection,

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}). \quad (4.16)$$

4.2 Parallel Transport and Geodesic

The directional derivative is appropriately covariantized:

$$\frac{d}{d\lambda} = \frac{dx^\mu}{d\lambda} \partial_\mu \quad \implies \quad \frac{D}{d\lambda} \equiv \frac{dx^\mu}{d\lambda} \nabla_\mu. \quad (4.17)$$

We can use this to define the concept of the *parallel transport* of a tensor along a path $x^\mu(\lambda)$ to be that the tensor is as covariantly constant as possible along the path:

$$\frac{D}{d\lambda} T = 0. \quad (4.18)$$

For a vector, this reads

$$\frac{d}{d\lambda} V^\rho + \Gamma^\rho_{\mu\nu} \frac{dx^\mu}{d\lambda} V^\nu = 0. \quad (4.19)$$

You can easily show that if two vectors are parallel transported, then their inner product with each other is covariantly constant along the curve, as long as the connection is metric compatible. Crucially, parallel transport with respect to a metric-compatible connection preserves vector norms and orthogonality.

Next, the *geodesic* is defined to be a path whose proper-time tangent vector $\frac{dx^\mu}{d\tau}$ is parallel transported along the path itself. This gives the geodesic equation,

$$\frac{d^2 x^\rho}{d\tau^2} + \Gamma^\rho_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0. \quad (4.20)$$

A geodesic between two points also minimizes the distance between those two points. For spacetime, that means maximize the proper time. You will investigate this way of defining a geodesic in the problem set. It's interesting that we have two seemingly independent ways to define a geodesic and that these two definitions are equivalent, as you will (partly) show in your problem set.

Let the four-velocity be

$$U^\mu = \frac{dx^\mu}{d\tau}. \quad (4.21)$$

For a timelike path, this is the momentum per unit mass. Then, for timelike paths, the geodesic equation may be written as

$$U^\nu \nabla_\nu U^\mu = 0. \quad (4.22)$$

This intuitively states that a particle experiencing zero force (free particle) continues to move in the same direction as its original velocity. The concept of “same direction” involves a covariant derivative, of course. That is, “covariantly” the same direction.

An affine parameter λ is defined to be one which is linear in the proper time:

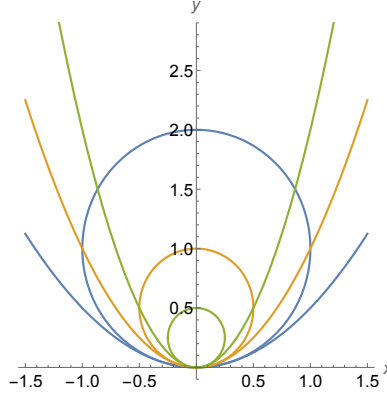
$$\lambda = a\tau + b. \quad (4.23)$$

The geodesic equation is preserved under a reparametrization from τ to λ , as long as λ is an affine parameter. If λ is an affine, then a “fictitious” force will appear on the RHS of the geodesic equation involving the acceleration of the parameter λ with respect to τ :

$$\frac{d^2 x^\rho}{d\lambda^2} + \Gamma^\rho_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = \frac{dx^\rho}{d\lambda} \frac{d}{d\tau} \left[\left(\frac{d\lambda}{d\tau} \right)^{-1} \right]. \quad (4.24)$$

4.3 Curvature in Low Dimensions

Consider a one-dimensional parabola, $y = \frac{1}{2}a\left(\frac{x}{a}\right)^2 = \frac{1}{2a}x^2$, where a is a constant with units of distance. You can “fit” a circle of radius a at the bottom of the parabola, as shown below:



The blue plot has $a = 1$, the orange has $a = \frac{1}{2}$, and the green has $a = \frac{1}{4}$.

This means that if we expand the circle around the bottom point to lowest order, then we will get the corresponding parabola. Let's check this. Note that the center of the circle is at the point $(x, y) = (0, a)$. This circle is, therefore, given by

$$y = a \pm \sqrt{a^2 - x^2}. \quad (4.25)$$

The bottom of the circle is at $x \approx 0$, so that $x \ll a$, and on the lower semicircle, i.e. with the minus sign in the above equation. Thus,

$$y \approx a - a \left[1 - \frac{1}{2} \left(\frac{x}{a} \right)^2 \right] = \frac{1}{2a} x^2, \quad (4.26)$$

as promised. The *inverse* of the radius of this circle is a measure of the degree of curvature of the parabola at its vertex: the smaller the circle, the more curved, the larger the circle, the more flat. This generalizes to any point on any smooth curve: you can always “fit” a circle tangent to the curve at the desired point such that when you expand the circle around that same point, you reproduce the curve itself to leading order. The only technicality left is the sign of the curvature. We have to make a choice of orientation as to what counts as “positive” curvature and what counts as “negative” curvature. We have to pick a direction in which to traverse the curve. Once we do that, then we have to decide whether a curve that is deflecting to the left as we move forward is positively or negatively curved, and the opposite for a curve that is deflecting to the right instead. For example, if we traverse the parabolas above from left to right (increasing x), then hopefully it's clear that, as you go through the vertex of the parabola, the curve is deflecting to your left. We might call that *positive* curvature. With that convention, an up-side-down parabola (one with a negative coefficient of the quadratic term, would have *negative* curvature. Convince yourself that this would mean that a sine or cosine curve is positively curved at its valleys and negatively curved at its peaks.

Let's do the same thing in 2d. Imagine the volume of revolution of the above plots around the y -axis, except now the vertical axis is the z -axis instead. We would then have a sphere given by the equation

$$x^2 + y^2 + (z - a)^2 = a^2. \quad (4.27)$$

Let's expand around the bottom:

$$z = a - \sqrt{a^2 - x^2 - y^2} \approx \frac{1}{2a} (x^2 + y^2). \quad (4.28)$$

So, the story is more or less the same as in 1d. The only difference is that we have one parabola for the x and one for the y direction. In fact, we will always have one for each direction on the manifold,

so n of them, if n is the dimension of the manifold. In the 2d case, in fact, the above result is not the most general situation. The most general situation is that

$$z \approx \frac{1}{2a}x^2 + \frac{1}{2b}y^2 + \frac{1}{c}xy = \frac{1}{2}(x, y) \begin{pmatrix} \frac{1}{a} & \frac{1}{c} \\ \frac{1}{c} & \frac{1}{b} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{2}x_i M^i_j x^j. \quad (4.29)$$

You can, of course, diagonalize the matrix M so that

$$z \approx \frac{1}{2a'}x'^2 + \frac{1}{2b'}y'^2, \quad (4.30)$$

where a' and b' are sum functions of a , b , and c , and x and y are some linear combinations of x and y with coefficients that may depend on a , b , and c . That is, we can always find coordinates x' and y' such that the form of the surface near a point on that surface is a parabola in the x' direction with radius of curvature equal to some a' , and in the y' direction with a radius of curvature equal to some b' .

In this 2d case, the situation is rather simple because there are only two coordinate-independent invariant quantities that you can construct from a 2×2 matrix: the determinant and the trace. The determinant turns out to be an *intrinsic* measure of curvature, completely independent of how might want to embed your manifold into a higher-dimensional manifold (e.g., a flat space), and the trace is *extrinsic*:

intrinsic curvature: $\det M = \frac{1}{a'b'} = \frac{1}{ab} - \frac{1}{c^2},$ extrinsic curvature: $(\frac{1}{2} \text{tr } M)^2 = \frac{1}{4}(\frac{1}{a'} + \frac{1}{b'})^2 = \frac{1}{4}(\frac{1}{a} + \frac{1}{b})^2.$	(4.31)
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The reason why the trace is squared is so that it has the same dimension as the determinant. The reason for putting the factor of $\frac{1}{4}$ there is so that the intrinsic and extrinsic curvatures of the sphere are equal. For the 2-sphere, S^2 , you should find $a = b$ and $c = 0$.

Exercise: Consider the surface $z = xy$ near the origin, $(x, y) = (0, 0)$. Show that the intrinsic and extrinsic curvatures are both -1 at the origin.

Exercise: Where on the torus does the curvature vanish?

4.4 Local Flatness

A manifold is supposed to be locally flat because there is always a neighborhood of a point that we can map to an neighborhood in flat space. Therefore, it follows that we can always choose coordinates near a point such that the metric is flat when expanded near that point. We might as well set the point to be the origin: $x = 0$. Thus, we are just Taylor-expanding the metric around $x = 0$:

$$g_{\mu\nu}(x) = g_{\mu\nu}(0) + (A_{\mu\nu})_\rho X^\rho + (B_{\mu\nu})_{\rho\lambda} x^\rho x^\lambda + \dots \quad (4.32)$$

Change coordinates according to

$$x^\mu = K^\mu_\nu x'^\nu + L^\mu_{\nu\rho} x'^\nu x'^\rho + M^\mu_{\nu\rho\lambda} x'^\nu x'^\rho x'^\lambda + \dots, \quad (4.33)$$

so that

$$\frac{\partial x^\mu}{\partial x'^\nu} = K^\mu_\nu + 2L^\mu_{(\nu\rho)} x'^\rho + 3!M^\mu_{(\nu\rho\lambda)} x'^\rho x'^\lambda + \dots \quad (4.34)$$

At the origin, the transformed metric is

$$g'_{\mu\nu}(0) = g_{\rho\lambda}(0) K^\rho_\mu K^\lambda_\nu. \quad (4.35)$$

This just represents a linear change of basis. There is always a choice of K that will diagonalize $g_{\rho\lambda}(0)$. Therefore, we could simply rescale coordinates such that the diagonal entries of the diagonalized matrix are just 1, and the final metric is just the identity matrix: $g_{\mu\nu}(0) = \delta_{\mu\nu}$. Note that we will keep dropping primes once we've gotten to the form of the metric that we want.

Note that K contains n^2 real components. We had to use $\frac{1}{2}n(n+1)$ of those to adjust $g_{\mu\nu}(0)$, which contains $\frac{1}{2}n(n+1)$ independent components, to be equal to $\delta_{\mu\nu}$. Therefore, the number of components in K is really

$$|K| = n^2 - \frac{1}{2}n(n+1) = \frac{1}{2}n(n-1). \quad (4.36)$$

Notice that that is exactly the number of rotations in n -dimensional space (i.e., it's the number of ways of picking two directions out of n directions to be the plane on which we rotate. This is not a surprise because this is precisely the number of transformations that keep $\delta_{\mu\nu}$ fixed.

Now, we move to the linear terms:

$$g_{\mu\nu}(x) = \delta_{\mu\nu} + (A_{\mu\nu})_{\rho} x^{\rho} + \dots \quad (4.37)$$

Change coordinates to

$$x^{\mu} = x'^{\mu} + L^{\mu}_{\nu\rho} x'^{\nu} x'^{\rho} + \dots \quad (4.38)$$

Note that $(A_{\mu\nu})_{\rho}$ contains $\frac{1}{2}n^2(n+1)$ elements because it has three indices, but it must be symmetric in 2 of them. But, that's exactly the same for $L^{\mu}_{\nu\rho}$: it has three indices and it must be symmetric in ν and ρ . Therefore, it also has $\frac{1}{2}n^2(n+1)$ elements. Thus, we have just enough freedom in choosing $L^{\mu}_{\nu\rho}$ to remove all of the $(A_{\mu\nu})_{\rho}$. Now,

$$g_{\mu\nu}(x) = \delta_{\mu\nu} + (B_{\mu\nu})_{\rho\lambda} x^{\rho} x^{\lambda} + \dots \quad (4.39)$$

The number of elements in B is

$$|(B_{\mu\nu})_{\rho\lambda}| = \left(\frac{1}{2}n(n+1)\right)^2. \quad (4.40)$$

Change coordinates to

$$x^{\mu} = x'^{\mu} + M^{\mu}_{\nu\rho\lambda} x'^{\nu} x'^{\rho} x'^{\lambda} + \dots \quad (4.41)$$

How many elements does M have?:

$$|M^{\mu}_{\nu\rho\lambda}| = \frac{1}{6}n^2(n+1)(n+2). \quad (4.42)$$

Thus, the number of components of B remaining is

$$\frac{1}{4}n^2(n+1)^2 - \frac{1}{6}n^2(n+1)(n+2) = \frac{1}{12}n^2(n^2-1). \quad (4.43)$$

These remaining components *are* the components of curvature; they are the obstructions to ironing the manifold out flat. This number is precisely the number of independent elements of the Riemann tensor. For example, last class, we learned that if $n = 2$, then the Riemann tensor contains only 1 independent component. Indeed, $\frac{1}{12}n^2(n^2-1) = 1$ when $n = 2$. The number of components in $n = 3$ is 6 and in $n = 4$ (relevant for the usual spacetime) is 20.

Chapter 5

Gravitation

Recall that the gravitational potential due to a point mass of mass M in Newtonian gravity is given by

$$\Phi = -\frac{GM}{r}, \quad (5.1)$$

where G is Newton's constant and r is the radial distance away from the point mass. By Gauss' law, this continues to be the potential everywhere *outside* a spherically symmetric mass distribution. In general, of course, the gravitational potential Φ generated by a mass density ρ is given by Poisson's equation

$$\nabla^2 \Phi = 4\pi G\rho. \quad (5.2)$$

Furthermore, the *gravitational field*, $\mathbf{g}$, experienced by an object in the presence of a gravitational potential is

$$\mathbf{g} = -\nabla\Phi. \quad (5.3)$$

You would multiply this field by the *gravitational mass*, m_G , of the object to get the gravitational force that it experiences, much the same way you would multiply the electric field by the electric charge of an object to get the electric force that it experiences. To translate that into the actual physical motion of the object, you would divide by the *inertial mass*, m_I , of the object:

$$m_I \frac{d^2 \mathbf{x}}{dt^2} = m_G \mathbf{g} \quad \implies \quad \frac{d^2 \mathbf{x}}{dt^2} = \frac{m_G}{m_I} \mathbf{g}. \quad (5.4)$$

Note, therefore, that the gravitational field $\mathbf{g}$ has units of acceleration. The gravitational mass is the analog of electric charge: it measures the strength of the (gravitation) force experienced by an object in a gravitational field. The inertial mass, on the other hand, is the mass that shows up in Newton's second law: it measures the resistance of an object to motion due to an external force. The latter does not care what type of force we're talking about; the inertial mass would be the same regardless of whether or not the force is electric, or magnetic, or gravitational, or of any other nature. The miracle is that $m_G = m_I$.

5.1 The Equivalence Principle

Since $m_G = m_I$ independent of the nature of the object feeling the gravitational force, we are (I mean, Einstein was) led naturally to the equivalence principle. This is Einstein's famous thought experiment of the freely falling person inside a freely falling elevator. Provided that the elevator is sufficiently small and we only consider it over relatively short periods of time, everything inside the elevator falls at the same rate as everything else, including the elevator itself. Therefore, with no external point of

reference, the free-falling observer inside the elevator cannot possibly know if they are falling in the presence of a gravitational field or not.

Strictly speaking, this is true only for a gravitational field produced by an infinite sheet of constant mass density. For the same reason as a constant charge density sheet produces a constant electric field, so the constant mass density sheet produces a constant gravitational field. The gravitational field of the Earth, however, is not constant: it points radially inwards and goes like the inverse square of the radial distance from the center of the Earth. For this reason, there will be a slight “squeezing” inwards towards the bottom of the elevator. From the perspective of the center of the elevator, in fact, light dust particles near the bottom of the elevator will be moving down and the dust particles closer to the top of the elevator will be moving upwards slightly. These are due to so-called *tidal forces*.

This is why we made the caveat of the elevator being small and only considering its fall (or non-fall, depending on your perspective) for a short period of time. Given this caveat, the equivalence principle states that a freely falling (non-rotating) laboratory is governed by the same laws of physics as those of special relativity. In particular, it must be possible to find locally inertial coordinates such that $g_{\mu\nu} = \eta_{\mu\nu}$ and $\partial_\rho g_{\mu\nu} = 0$ at a particular point on the spacetime manifold.

5.2 The Principle of Minimal Coupling

This leads us to a powerful recipe for bootstrapping the physics of curved spacetime to that of flat spacetime, called the *principle of minimal coupling*: in any law of physics valid in flat spacetime, simply replace the flat Minkowski metric with a general metric and the partial derivatives with covariant ones. Et voila, you have the curved spacetime version of that physical law. For example, straight lines in flat space are defined as embeddings $x^\mu(\lambda)$ that satisfy the “zero-acceleration” rule

$$\frac{d^2 x^\mu}{d\lambda^2} = 0 \quad \implies \quad \frac{dx^\nu}{d\lambda} \partial_\nu \frac{dx^\mu}{d\lambda} = 0. \quad (5.5)$$

The principle of minimal coupling dictates that the generalization of this to curved space is

$$\frac{dx^\nu}{d\lambda} \partial_\nu \frac{dx^\mu}{d\lambda} \rightarrow \frac{dx^\nu}{d\lambda} \nabla_\nu \frac{dx^\mu}{d\lambda} = \frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\nu\rho} \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda}, \quad (5.6)$$

which immediately gives the geodesic equation. Therefore, the geodesic is the curved spacetime analog of a straight line. In other words, Newton’s first law, which states that an object at rest will remain at rest and an object moving at constant velocity will continue traveling in a straight line at the constant velocity unless acted upon by an imbalanced force, translates to geodesics in curved spacetime.

5.3 Weak-field Limit of Gravity

The Newtonian limit of relativistic gravity will be the following:

1. Slowly-moving particles,

$$\frac{dx^i}{d\tau} \ll \frac{dt}{d\tau}. \quad (5.7)$$

2. Weak gravitational field,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (5.8)$$

3. Static gravitational field,

$$\partial_0 h_{\mu\nu} = 0. \quad (5.9)$$

Point 1 above implies that the geodesic equation becomes

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} \left(\frac{dt}{d\tau} \right)^2 = 0, \quad (5.10)$$

with

$$\Gamma^\mu_{00} = -\frac{1}{2} g^{\mu\nu} \partial_\nu g_{00}. \quad (5.11)$$

In addition, Point 2 above implies

$$\Gamma^\mu_{00} \approx -\frac{1}{2} \eta^{\mu\nu} \partial_\nu h_{00} \quad \implies \quad \frac{d^2 x^\mu}{d\tau^2} = \frac{1}{2} \eta^{\mu\nu} \partial_\nu h_{00} \left(\frac{dt}{d\tau} \right)^2, \quad (5.12)$$

where use was made of the fact that

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + O(h^2), \quad (5.13)$$

where $h^{\mu\nu} = \eta^{\mu\rho} \eta^{\nu\lambda} h_{\rho\lambda}$, and that we drop anything quadratic and higher in the perturbation h .

Since $\partial_0 h_{00} = 0$, the $\mu = 0$ component of the geodesic equation just implies that t is an affine parameter,

$$\frac{d^2 t}{d\tau^2} = 0. \quad (5.14)$$

The spatial components of the geodesic equation read

$$\frac{d^2 \mathbf{x}}{d\tau^2} = -\frac{1}{2} \left(\frac{dt}{d\tau} \right)^2 \nabla h_{00}. \quad (5.15)$$

Since t is an affine parameter, we can simply write the above equation as

$$\frac{d^2 \mathbf{x}}{dt^2} = -\frac{1}{2} \nabla h_{00}. \quad (5.16)$$

Compare this with

$$\frac{d^2 \mathbf{x}}{dt^2} = -\nabla \Phi, \quad (5.17)$$

to conclude that

$$h_{00} = 2\Phi. \quad (5.18)$$

Thus, the metric is approximately

$$g_{00} = 1 + 2\Phi, \quad g_{0i} = g_{i0} = 0, \quad g_{ij} = -\delta_{ij}. \quad (5.19)$$

Evidently, Φ must be dimensionless. To make (5.1) dimensionless, we must divide by c^2 :

$$\Phi = -\frac{GM}{c^2 r}. \quad (5.20)$$

I prefer to write this in terms of fundamental, Planck units, via the Planck mass and Planck length,

$$M_P = \sqrt{\frac{\hbar c}{G}} = 2.176 \times 10^{-8} \text{ kg}, \quad \ell_P = \sqrt{G\hbar c^3} = 1.616 \times 10^{-35} \text{ m}, \quad (5.21)$$

so that

$$\Phi = -\frac{M/M_P}{r/\ell_P}. \quad (5.22)$$

The typical length scales and mass scales of interest to us are much much bigger than the Planck length and Planck mass, respectively. However, the effect in length is usually much greater than that for mass. Therefore, indeed, $|\Phi| \ll 1$ for most intents and purposes. For example, for the surface of the earth, $M = 5.97 \times 10^{24} \text{ kg}$ and $r = 6.378 \times 10^6 \text{ m}$ so that $M/M_P \approx 10^{32}$, whereas $r/\ell_P = 10^{41}$ and so $|\Phi| = 10^{-9}$. The mass of the Sun is about a 3.33×10^5 times bigger than that of Earth, but its radius is only 100 times bigger. Thus, on the surface of the Sun, $|\Phi| \approx 10^{-6}$. A solar mass white dwarf has a radius on the order of the radius of the Earth. Thus, $|\Phi| \approx 10^{-4}$.

Chapter 6

Gravitational Field Equations

6.1 Energy-Momentum Tensor

Consider a gas of particles each of proper mass m_0 and number density n_0 in a frame in which the gas as a whole has zero net velocity. In this frame, the mass density would be $\rho_0 = m_0 n_0$. Then, boost this system. Because of length contraction, we know that the volume occupied by this same number of gas particles will be smaller and so the density will go up. Furthermore, the energy of the gas particles will be γm , where γ is the gamma factor associated with the boost. Thus, the number density increases by a factor of γ and the energy per particle increases by a factor of the γ , and so ρ increases by a factor of γ^2 . No matter what, we see that ρ is *not* a scalar quantity. In fact, because it gains two factors of γ , we know that it has to be the 00-component of a (2, 0)-tensor:

$$T^{\mu\nu} = \rho u^\mu u^\nu, \quad (6.1)$$

where u^μ is the fluid 4-velocity. In a local Cartesian inertial frame near any particular point, we can always write u^μ as $\gamma_u(1, \mathbf{u})$.

Looking at the components in a local Cartesian inertial frame, we can interpret the components of the energy-momentum tensor:

- T^{00} is the energy density;
- T^{0i} is the energy flux in i -direction;
- T^{i0} is the momentum density in the i -direction;
- T^{ij} is the flux of the i -component of momentum per unit area in the j -direction.

Note that there is absolutely no a-priori reason why T^{0i} should be equal to T^{i0} . This is a special consequence of invariance under Lorentz boosts. If you're interested in what happens without boost invariance, you can take a look at the work of Jan de Boer et al. on perfect fluids without boost-invariance: <https://arxiv.org/abs/1710.04708>.

A perfect fluid is defined to be one for which there are no forces between the partricles, and no heat conduction or viscosity in the frame which is comoving with the fluid (the instantanenous rest frame, or IRF). In your problem set, you are asked to show that this means that the EMT in the IRF reads

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu - p\eta^{\mu\nu}. \quad (6.2)$$

This is flat spacetime. In the spirit of minimal coupling, all we would do to covariantize the above expression is change $\eta^{\mu\nu}$ into $g^{\mu\nu}$. A common special case is called presureless dust, where we set $p = 0$.

In fact, the energy-momentum tensor ought to be the Noether current associated with spacetime-translation symmetry. In that context, it naturally appears with one upper index and one lower index: the lower index corresponds to the spacetime direction in which we are translating and the upper index are the spacetime *components* of the current associated with that translation. Then, translation symmetry implies that this Noether current must be conserved: $\partial_\mu T^{\mu\nu} = 0$. Again, that's in flat spacetime. Minimal coupling dictates that it should be generalized into

$$\nabla_\mu T^{\mu\nu} = 0. \quad (6.3)$$

If we write this in flat spacetime, for a perfect fluid, and take the non-relativistic limit, we will get

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (6.4)$$

which is the classical continuity equation for a fluid.

Plugging this back into the energy-momentum conservation equation and taking the limit of small pressure, you can derive the famous Euler equation,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla p}{\rho}. \quad (6.5)$$

It's not much more work to get from here to the theory of more complicated fluids. For example, the Toner-Tu theory of flocks is described by a modified Euler equation of the form

$$\frac{\partial \mathbf{u}}{\partial t} + \lambda_1 (\mathbf{u} \cdot \nabla) \mathbf{u} + \lambda_2 (\nabla \cdot \mathbf{u}) \mathbf{u} + \lambda_3 \nabla(|\mathbf{u}|^2) = (\alpha - \beta |\mathbf{u}|^2) \mathbf{u} - \frac{\nabla p}{\rho} + \dots, \quad (6.6)$$

where $\dots$ contains second-derivative “diffusion” terms like $D_T \nabla^2 \mathbf{u}$ and $D_N \nabla(\nabla \cdot \mathbf{u})$, as well as Gaussian white-noise forces that are meant to model thermal fluctuations, or, indeed, any other random processes.

6.2 Einstein Equations

We've already seen that if the gravitational field is weak, then the Newton potential enters the metric via $g_{00} = 1 + 2\Phi$. Therefore, from the Poisson equation of Newtonian gravity, we derive

$$\nabla^2 \Phi = 4\pi G \rho \quad \implies \quad \nabla^2 g_{00} = 8\pi G \rho. \quad (6.7)$$

Let's consider pressureless dust, for which

$$T_{00} = \rho, \quad (6.8)$$

so that

$$\nabla^2 g_{00} = 8\pi G T_{00}. \quad (6.9)$$

Of course, this has to be promoted to a tensor equation. The RHS is already suggestive of $8\pi G T_{\mu\nu}$. The LHS would have to be a $(2, 0)$ -tensor which is a generalized second derivative of the metric. That's sounds exactly like a curvature, for example the Ricci tensor. Assuming that the energy-momentum tensor *is* symmetric, $T_{\mu\nu} = T_{\nu\mu}$, which, as I've already said, requires boost-invariance, the LHS must also be symmetric. In fact, the most general term we could have here is

$$a R_{\mu\nu} + b R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad \kappa = 8\pi G. \quad (6.10)$$

Let's first consider the case $\Lambda = 0$. You can show that in order for the covariant divergence of the LHS to vanish, we would have to have $b = -a/2$. Finally, the value of a is determined by ensuring that the equation has the correct Newtonian limit: $a = -1$. Thus, we arrive at Einstein's field equations,

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -\kappa T_{\mu\nu}. \quad (6.11)$$

The LHS tensor is called the Einstein tensor,

$$E_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R. \quad (6.12)$$

You can show that Einstein's equation can be written equivalently as

$$R_{\mu\nu} = -\kappa(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}), \quad (6.13)$$

where $T = g^{\mu\nu}T_{\mu\nu}$.

This promotes the one Newton's field equation into 10 independent equations! Furthermore, Newton's field equation is linear in Φ , whereas Einstein's equations are quite non-linear in $g_{\mu\nu}$. In general, these are very difficult equations to solve.

Remember that the Riemann tensor contains 20 degrees of freedom in 4 spacetime dimensions. Therefore, even if we were in complete vacuum, where $T_{\mu\nu} = 0$, and so $R_{\mu\nu} = 0$, that would still not imply that spacetime would have to be flat, because there are 10 more components of the Riemann tensor that may still be nonzero. Compare this with 3-dimensional spacetime, where the number of components of the Riemann tensor is 6 and the number of field equations is also 6, and so if the Ricci tensor vanishes, then so does the Riemann tensor! Therefore, the only solution to vacuum Einstein's equations in the case of 3-dimensional spacetime is flat spacetime.

6.3 Cosmological Constant

Putting back the cosmological constant term gives

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = -\kappa T_{\mu\nu}, \quad (6.14)$$

or

$$R_{\mu\nu} = -\kappa(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}) + \Lambda g_{\mu\nu}. \quad (6.15)$$

The Newtonian limit of this is

$$\nabla^2\Phi = 4\pi G\rho - \Lambda, \quad (6.16)$$

and, for a spherical mass M , Gauss' law would give

$$\mathbf{g} = -\nabla\Phi = \left(-\frac{GM}{r^2} + \frac{\Lambda r}{3}\right)\hat{\mathbf{r}}. \quad (6.17)$$

So, if the cosmological constant is negative, then it acts just like an ideal spring. On the other hand, if $\Lambda > 0$, then it is like a "reverse" Hooke's law that tends to push apart rather than restore.

Without the cosmological constant, Einstein was unable to construct static solutions to the field equations. He added the cosmological constant in order to do just that. Why? Because, at the time, in 1916, people thought the Milky Way Galaxy was the entire universe that was static. Only later were many more galaxies discovered and in 1929 Edwin Hubble made his groundbreaking discovery of the expansion of the universe by measuring the redshifts of those galaxies. So, it turned out that the universe was *not* static, so the cosmological constant was not needed after all.

Actually, the cosmological constant saga was not over. Dark energy was later discovered in 1998 when astronomers led by Adam Riess, Saul Perlmutter, and Brian Schmidt discovered that the universe expansion rate was in fact increasing! The standard picture for dark energy is that it *is* the cosmological constant. This is part of the so-called Λ -CDM standard model of cosmology (Λ stands for the cosmological constant, which constitutes dark energy, and CDM stands for cold dark matter). The cosmological constant can instead be placed on the RHS of the Einstein field equations and interpreted as part of the energy-momentum tensor. However, it's a strange type of substance that satisfies $p = -\rho$,

with negative pressure! Indeed, if it is positive, it tends to push things apart, as we already noted. It imbues vacuum spacetime itself with an energy density

$$\rho_{\text{vac}} = \frac{\Lambda}{8\pi G}. \quad (6.18)$$

So, if we define the vacuum energy-momentum tensor

$$T_{\mu\nu}^{\text{vac}} = \rho_{\text{vac}} g_{\mu\nu}, \quad (6.19)$$

then Einstein's equations read

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -\kappa (T_{\mu\nu} + T_{\mu\nu}^{\text{vac}}). \quad (6.20)$$

For this discovery, those three scientists won the 2011 Nobel Prize. We had lots of parties and celebrations at UC Berkeley, as I was a grad student at the time.

Actually, the cosmological constant constitutes the largest naturalness problem. The question is why is it so small in fundamental units, 10^{-123} per Planck area? Usually, if something is so fundamentally small, there should be a good symmetry reason for it: if the symmetry were *exact*, then that quantity would be *exactly* zero. It is only nonzero because the symmetry is slightly broken. But no such symmetry has been proposed or discovered. The classic text that explores this so-called Cosmological Constant Problem is by Cliff Burgess: <https://arxiv.org/abs/1309.4133>.

6.4 Variational Principle

We seek an actional principle whose Euler-Lagrange equations of motion produce the Einstein field equations. A putative action for gravity would have to take the form $S \sim \int d^4x \sqrt{-g} \mathcal{L}$, where $\mathcal{L}$ is a Lagrangian density, which must be a scalar quantity. As usual, the Lagrangian density should be a functional of the metric and its derivatives. To leading order, as usual, we presume that it only depends on the first derivative. The correct generalized first derivative would be the connection itself. There really isn't much we can put in $\mathcal{L}$ at lowest order in derivatives:

$$\mathcal{L} = aR + b, \quad (6.21)$$

where R is the Ricci scalar and a and b are constants.

So, now, we just have to vary this action with respect to the metric:

$$g_{\mu\nu}(x) \rightarrow g'_{\mu\nu}(x) = g_{\mu\nu}(x) + \delta g_{\mu\nu}(x). \quad (6.22)$$

We think of $\delta g_{\mu\nu}$ as infinitesimal. If you want, you can put an ϵ in front of it, keep only up to lowest non-trivial order in ϵ and then set $\epsilon = 1$ at the end. Note that it is important that we are considering the new metric at the *same* or *old* spacetime point. So, this transformation contains not only the tensorial transformation of the metric, but also a ‘Taylor expansion’ part as we have to move from the new point x' to the old point x .

First, let us work out the transformation of the determinant of the metric. Use the important relationship

$$\ln \det = \text{tr} \ln. \quad (6.23)$$

You can prove this by first noting that both $\det$ and tr are left unchanged by a change of basis and thus by the process of diagonalizing a matrix. Therefore, without loss of generality, we only need to show that the above relation holds for a diagonal matrix, for which it is obvious. Note that the log of

a diagonal matrix is just the diagonal matrix whose diagonal elements are the log of the corresponding diagonal element of the original diagonal matrix.

In spacetime, because of the signature, we should put a minus sign in front of the determinant. Denote by $\mathbf{g}$ the spacetime metric as a matrix. Then,

$$\ln(-\det(\mathbf{g})) = \text{tr} \ln(\mathbf{g}), \quad (6.24)$$

so that taking a variation reads

$$\frac{\delta(-\det(\mathbf{g}))}{-\det(\mathbf{g})} = \text{tr}[\mathbf{g}^{-1}\delta\mathbf{g}] = g^{\mu\nu}\delta g_{\mu\nu}. \quad (6.25)$$

Therefore, denoting $-\det(\mathbf{g}) = -g$ again, we find

$$\delta(-g) = (-g)g^{\mu\nu}\delta g_{\mu\nu}, \quad (6.26)$$

and so

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g}g^{\mu\nu}\delta g_{\mu\nu}. \quad (6.27)$$

From the relation $g^{\mu\rho}g_{\rho\nu} = \delta^\mu_\nu$, you should be able to show that

$$\delta g^{\mu\nu} = -g^{\mu\rho}g^{\nu\lambda}\delta g_{\rho\lambda}, \quad (6.28)$$

and so, we could equally well write

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}. \quad (6.29)$$

Next, we need to take the variation of $\mathcal{L}$, which we said is just $aR + b$. Therefore, we need the variation of the Ricci scalar. Let's start with the Christoffel symbols. First, define

$$\Gamma_{\lambda\mu\nu} = \frac{1}{2}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}), \quad (6.30)$$

such that

$$\Gamma^\rho{}_{\mu\nu} = g^{\rho\lambda}\Gamma_{\lambda\mu\nu}. \quad (6.31)$$

Thus,

$$\delta\Gamma^\rho{}_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\partial_\mu\delta g_{\lambda\nu} + \partial_\nu\delta g_{\mu\lambda} - \partial_\lambda\delta g_{\mu\nu}) + \frac{1}{2}\delta g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}). \quad (6.32)$$

You are asked in the problem set to show that this is the same as

$$\delta\Gamma^\rho{}_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\nabla_\mu\delta g_{\lambda\nu} + \nabla_\nu\delta g_{\mu\lambda} - \nabla_\lambda\delta g_{\mu\nu}). \quad (6.33)$$

In the problem set you are also asked to find the variation of the Ricci tensor and the Ricci scalar. The variation of the Ricci tensor involves covariant derivatives of the variation of the Christoffel symbols. Bear in mind that the Christoffel symbols do not form a tensor, but their variation does! Therefore, you can take the covariant derivative of the variation of the Christoffel symbols exactly as you would a tensor. Eventually, you should end up with the following result:

$$\delta[\sqrt{-g}(R - 2\Lambda)] = \sqrt{-g}(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} + g_{\mu\nu}\square - \nabla_\mu\nabla_\nu)\delta g^{\mu\nu}. \quad (6.34)$$

Therefore, other than the overall factor of $\sqrt{-g}$, the coefficient of $\delta g^{\mu\nu}$ is precisely the expression that shows up on the LHS of Einstein's equations *except* for the derivatives acting on $\delta g^{\mu\nu}$. We give the Lagrangian a name: the Einstein-Hilbert Lagrangian,

$$\mathcal{L} = R - 2\Lambda, \quad (6.35)$$

and its integral over spacetime is the Einstein-Hilbert action.

Of course, the variation (6.34) is supposed to be integrated over d^4x , so the last two total derivative terms actually end up being boundary terms (by the divergence theorem). They don't actually contribute to the equations of motion. Nevertheless, we do have to take care of them. Those total derivative terms are handled by adding a boundary term to the Einstein-Hilbert action called the Gibbons-Hawking-York (GHY) term. The whole point of the GHY term is that its variation exactly cancels the total derivative terms in (6.34). You can look this term up on Wikipedia, but I will not discuss it further, at least for now.

6.5 RHS of Einstein's Equations

The RHS of Einstein's equations contain the energy-momentum tensor of the "matter" (anything other than just pure gravity) that is sitting in spacetime. Let's consider the simplest example of a real massive scalar field. Let's first consider it in flat spacetime. The action is

$$S = \frac{1}{2} \int d^4x (\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2). \quad (6.36)$$

This is also called a real Klein-Gordon field. Take the variation of this:

$$\delta S = \int d^4x [\partial^\mu \phi \partial_\mu \delta \phi - m^2 \phi \delta \phi]. \quad (6.37)$$

Integrate by parts:

$$\delta S = - \int d^4x \delta \phi (\partial^2 + m^2) \phi, \quad (6.38)$$

where $\partial^2 = \partial_\mu \partial^\mu$. So the equation of motion is

$$(\partial^2 + m^2) \phi = 0. \quad (6.39)$$

Now, it should be obvious that this theory is invariant under spacetime translations:

$$x^\mu \rightarrow x'^\mu = x^\mu - \xi^\mu, \quad (6.40)$$

where ξ^μ is a constant. Under this transformation, a scalar field does not transform. That means the new scalar field ϕ' evaluated at the new point x' is the same as the old field ϕ evaluated at the old point x :

$$\phi'(x') = \phi(x). \quad (6.41)$$

But what is $\delta \phi$? First of all, $\delta \phi$ is a function: $\delta \phi(x)$. It's a function of *one* spacetime point. But, which is it? Is it x or x' ? Well, it doesn't actually matter, but, as a matter of convention, we usually pick the original (old) point. So,

$$\delta \phi(x) = \phi'(x) - \phi(x). \quad (6.42)$$

That is, it is the difference between the new field and the old field, but both evaluated at the old point. So, we need to find the new field evaluated at the old point. That's simple; it's just a Taylor expansion:

$$\phi'(x') = \phi'(x) + (x' - x)^\mu \partial_\mu \phi'(x) + \dots = \phi'(x) - \xi^\mu \partial_\mu \phi'(x) + \dots. \quad (6.43)$$

I will consider ξ^μ to be an infinitesimal translation and so I will only keep up to the lowest non-trivial order in ξ . The difference between ϕ' and ϕ at x is already first-order in ξ , so in $\partial_\mu \phi'(x)$, we can simply remove the primed symbol because that term is multiplied by ξ^μ and I'm only keeping up to first order in ξ . Thus,

$$\phi'(x') = \phi'(x) - \xi^\mu \partial_\mu \phi(x). \quad (6.44)$$

On the other hand, this is equal to $\phi(x)$, according to (6.41). Thus,

$$\delta\phi(x) = \phi'(x) - \phi(x) = \xi^\mu \partial_\mu \phi(x). \quad (6.45)$$

Plug this into (6.38):

$$\delta S = - \int d^4x \xi^\nu (\partial_\nu \phi) (\partial^2 + m^2) \phi. \quad (6.46)$$

The mass term, we can write as

$$(\partial_\nu \phi) m^2 \phi = \partial_\nu (\tfrac{1}{2} m^2 \phi^2). \quad (6.47)$$

For the other term, note that

$$\begin{aligned} \partial_\mu (\partial^\mu \phi \partial_\nu \phi) &= (\partial^2 \phi) \partial_\nu \phi + (\partial^\mu \phi) (\partial_\nu \partial_\mu \phi) \\ &= (\partial_\nu \phi) \partial^2 \phi + \partial_\nu (\tfrac{1}{2} \partial_\mu \phi \partial^\mu \phi), \end{aligned} \quad (6.48)$$

and so

$$(\partial_\nu \phi) \partial^2 \phi = \partial_\mu (\partial^\mu \phi \partial_\nu \phi) - \partial_\nu (\tfrac{1}{2} \partial_\mu \phi \partial^\mu \phi). \quad (6.49)$$

Plugging these results into (6.46) gives

$$\begin{aligned} \delta S &= - \int d^4x \xi^\nu [\partial_\mu (\partial^\mu \phi \partial_\nu \phi) - \partial_\nu (\tfrac{1}{2} \partial_\mu \phi \partial^\mu \phi - \tfrac{1}{2} m^2 \phi^2)] \\ &= - \int d^4x \xi^\nu \partial_\mu [\partial^\mu \phi \partial_\nu \phi - (\tfrac{1}{2} \partial_\mu \phi \partial^\mu \phi - \tfrac{1}{2} m^2 \phi^2) \delta_\nu^\mu]. \end{aligned} \quad (6.50)$$

If the transformation generated by ξ^ν is to be a symmetry of the theory, the above expression must vanish. Therefore, the derivative of the expression in the square brackets above must vanish. That's precisely the energy momentum conservation equation, where the energy-momentum tensor is

$$T^\mu{}_\nu = \partial^\mu \phi \partial_\nu \phi - (\tfrac{1}{2} \partial_\mu \phi \partial^\mu \phi - \tfrac{1}{2} m^2 \phi^2) \delta_\nu^\mu. \quad (6.51)$$

As I've already stated in lecture, the lower index ν labels which translation you're talking about. You know this because it is the index that is contracted with ξ^ν . In other words, it is the direction in which you are doing to the translation. The upper μ index on $T^\mu{}_\nu$ are the spacetime components of the associated current. That's the one which is hit by a derivative to give 0. In other words, energy-momentum conservation reads

$$\partial_\mu T^\mu{}_\nu = 0, \quad (6.52)$$

and there are four equations here: one for each direction ν in which we can do a translation. The energy-momentum tensor that you compute in this way is called the *canonical* energy-momentum tensor and is sometimes denoted by $\Theta^\mu{}_\nu$.

You can see that in this way of doing things, the upper index and the lower index on the energy-momentum tensor are really two complete different things. It's really rather a miracle that we should end up with something symmetric once one is raised or lowered:

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - (\tfrac{1}{2} \partial_\mu \phi \partial^\mu \phi - \tfrac{1}{2} m^2 \phi^2) \eta_{\mu\nu}. \quad (6.53)$$

Actually, we don't really know that this is what we would naturally call the energy-momentum tensor because if we were to multiply this by any number, we would still get a current which is conserved. So, how do I know that the overall coefficient is 1? The answer is that you just have to evaluate, for example T_{00} , and compare it with what you want it to be:

$$T_{00} = \dot{\phi}^2 - \tfrac{1}{2} [\dot{\phi}^2 - (\nabla \phi)^2 - m^2 \phi^2] = \dot{\phi}^2 + (\nabla \phi)^2 + m^2 \phi^2 = H, \quad (6.54)$$

where H is the Hamiltonian of a real massive scalar field. So, the overall coefficient is, indeed, 1.

How should this story change if the spacetime is not flat? Well, it's pretty obvious: $T_{\mu\nu}$ needs to be a tensor, so it should just turn into

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \left(\frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi - \frac{1}{2} m^2 \phi^2 \right) g_{\mu\nu}. \quad (6.55)$$

Actually, we don't need to turn the partial derivatives into covariant derivatives since they are identical when acting on a scalar field:

$$\nabla_\mu \phi = \partial_\mu \phi. \quad (6.56)$$

But, I'll just do it anyway so that we are reminded that this is on a curved background spacetime.

By minimal coupling, the action on a curved background is

$$S = \frac{1}{2} \int d^4x \sqrt{-g} (g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - m^2 \phi^2), \quad (6.57)$$

where I've been very careful to remind you that there is a factor of the inverse metric contracting the indices of the covariant derivatives acting on ϕ . The other way to compute an energy-momentum tensor is by taking the variation of the matter action with respect to the metric (or inverse metric):

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}}. \quad (6.58)$$

This is sometimes called the *Hilbert* energy-momentum tensor. It is obviously guaranteed to be symmetric, so, in general, it is different from the canonical one. Both are conserved, of course. You are asked to compute the Hilbert energy-momentum tensor for the real massive scalar field in your problem set and to compare your result with the canonical energy-momentum tensor (6.55). In the action, the only metric-dependent things that show up are the $\sqrt{-g}$ and $g^{\mu\nu}$ factors. I've already shown you how to relate the variations of those things with the variation of the metric, $\delta g_{\mu\nu}$. However, how do you vary $g_{\mu\nu}$ with respect to translations? That's easy: $g_{\mu\nu}$ is a tensor:

$$\begin{aligned} g'_{\mu\nu}(x') &= \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\lambda}{\partial x'^\nu} g_{\rho\lambda}(x) \\ &= \left(\delta_\mu^\rho + \frac{\partial \xi^\rho}{\partial x'^\mu} \right) \left(\delta_\nu^\lambda + \frac{\partial \xi^\lambda}{\partial x'^\nu} \right) g_{\rho\lambda}(x) \\ &= g_{\mu\nu}(x) + g_{\mu\rho}(x) \partial_\nu \xi^\rho(x) + g_{\rho\nu} \partial_\mu \xi^\rho(x) + O(\xi^2). \end{aligned} \quad (6.59)$$

Again, on the RHS, we only keep up to linear order in ξ and so we can ignore the difference between x and x' when taking the derivative or ξ . On the other hand,

$$g'_{\mu\nu}(x') = g'_{\mu\nu}(x) + (x' - x)^\rho \partial_\rho g'_{\mu\nu}(x) = g'_{\mu\nu}(x) - \xi^\rho \partial_\rho g_{\mu\nu}(x) + O(\xi^2). \quad (6.60)$$

Therefore,

$$\delta g_{\mu\nu} = \xi^\rho \partial_\rho g_{\mu\nu} + g_{\mu\rho} \partial_\nu \xi^\rho + g_{\rho\nu} \partial_\mu \xi^\rho. \quad (6.61)$$

I will leave it as an exercise for you to show that this is equivalent to

$$\delta g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu. \quad (6.62)$$

You can go through the whole derivation again. Now, you have to take into account not just the variation of the scalar field, but also the variation of the metric. Then, plug in the variations of the scalar field and the metric with respect to the translation of coordinates. The final result must look like

$$\delta S \propto \int d^4x \sqrt{-g} \xi^\nu \nabla_\mu (\text{something}). \quad (6.63)$$

The energy-momentum tensor is then proportional to the “something.”

Chapter 7

Schwarzschild

7.1 General Static Isotropic Metric

I want the most general rotationally-invariant metric. The angular part of the metric must be proportional to the metric on the 2-sphere, S^2 , which is $d\theta^2 + \sin^2 \theta d\phi^2$. So, the only possible pieces left over are dt^2 , $dt dr$ and dr^2 . Furthermore, the metric components themselves (i.e., the coefficients) cannot be a function of θ or ϕ . Therefore, the line element must look like

$$ds^2 = A(t, r)dt^2 - B(t, r)dtdr - C(t, r)dr^2 - D(t, r)(d\theta^2 + \sin^2 \theta d\phi^2). \quad (7.1)$$

I claim that, without loss of generality, we can actually assume that $D(t, r) = r^2$. This is because, we can always define a new radial coordinate $\bar{r}$ such that

$$\bar{r}^2 = D(t, r). \quad (7.2)$$

Then,

$$2\bar{r}d\bar{r} = \frac{\partial D}{\partial t}dt + \frac{\partial D}{\partial r}dr, \quad (7.3)$$

and

$$d\bar{r} = \frac{1}{2\sqrt{D}} \left(\frac{\partial D}{\partial t}dt + \frac{\partial D}{\partial r}dr \right) = \frac{\partial \sqrt{D}}{\partial t}dt + \frac{\partial \sqrt{D}}{\partial r}dr, \quad (7.4)$$

or

$$dr = \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-1} \left(d\bar{r} - \frac{\partial \sqrt{D}}{\partial t}dt \right), \quad (7.5)$$

and thus,

$$\begin{aligned} ds^2 &= A dt^2 - B dt \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-1} \left(d\bar{r} - \frac{\partial \sqrt{D}}{\partial t}dt \right) - C \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-2} \left(d\bar{r} - \frac{\partial \sqrt{D}}{\partial t}dt \right)^2 \\ &\quad - \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2) \\ &= A' dt^2 - B' dt d\bar{r} - C' d\bar{r}^2 - \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2), \end{aligned} \quad (7.6)$$

where

$$A' = A + B \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-1} \left(\frac{\partial \sqrt{D}}{\partial t} \right) - C \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-2} \left(\frac{\partial \sqrt{D}}{\partial t} \right)^2, \quad (7.7a)$$

$$B' = B \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-1} - 2C \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-2} \left(\frac{\partial \sqrt{D}}{\partial t} \right)^2, \quad (7.7b)$$

$$C' = C \left(\frac{\partial \sqrt{D}}{\partial r} \right)^{-2}. \quad (7.7c)$$

Well, A , B , and C are arbitrary functions, so A' , B' , and C' are equally arbitrary functions. So, we could have just started here. Therefore, we will just remove the primes on A' , B' , and C' and remove the bar on $\bar{r}$:

$$ds^2 = A dt^2 - B dt dr - C dr^2 - r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (7.8)$$

Next, I claim that, without loss of generality, we can assume that $B = 0$. This is because we can always define a new time coordinate $\bar{t}$ such that

$$d\bar{t} = \Phi(t, r)[A(t, r)dt - \frac{1}{2}B(t, r)dr]. \quad (7.9)$$

Here, Φ is an integrating factor whose purpose is to turn the RHS into a total derivative. In other words, we need

$$\Phi A = \frac{\partial \Psi}{\partial t}, \quad -\frac{1}{2}\Phi B = \frac{\partial \Psi}{\partial r}, \quad (7.10)$$

for *some* function Ψ . There are existence theorems that guarantee the existence of such a pair Φ and Ψ . They're not unique, of course, but that's fine. Let's not bother with the proofs. Proceeding, we get

$$d\bar{t}^2 = \Phi^2(A^2 dt^2 - AB dt dr + \frac{1}{4}B^2 dr^2), \quad (7.11)$$

and so

$$A dt^2 - B dt dr = \frac{1}{A\Phi^2} d\bar{t}^2 - \frac{B^2}{4A} dr^2, \quad (7.12)$$

and

$$ds^2 = \bar{A}(\bar{t}, r) d\bar{t}^2 - \bar{B}(\bar{t}, r) dr^2 - r^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad (7.13)$$

where

$$\bar{A} = \frac{1}{A\Phi^2}, \quad \bar{B} = C + \frac{B^2}{4A}. \quad (7.14)$$

Again, we can just remove the bars and start from here:

$$ds^2 = A(t, r) dt^2 - B(t, r) dr^2 - r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (7.15)$$

For the spacetime to be *stationary*, we need the metric to be independent of the time coordinate:

$$ds^2 = A(r) dt^2 - B(r) dr^2 - r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (7.16)$$

For the spacetime to be *static*, we need ds^2 to be invariant under time-reversal: $t \rightarrow -t$. Because it is already static and we were able to remove any piece which was linear in dt , the above spacetime is actually already static!

7.2 Einstein's Equations in Vacuum

In vacuum, Einstein's equations in trace-reversed form read

$$R_{\mu\nu} = 0. \quad (7.17)$$

You can compute the Ricci tensor manually. It's rather a long and tedious calculation. You can see this done in Hobson et al. I will just use Mathematica. (See Mathematica file). It happens that the Ricci tensor is actually diagonal. Furthermore, $R_{33} = R_{22} \sin^2 \theta$, and so R_{33} is not independent. Therefore, there are really only three equations: $R_{00} = 0$, $R_{11} = 0$, and $R_{22} = 0$. However, we are trying to find

two functions A and B . Therefore, this is an overcomplete set of equations. Generically, you might expect there to be *no* solution. However, you know that there has to be a solution. How can there be no isotropic and static metric for the spacetime outside a spherically symmetric mass distribution?

All of the components of the Ricci tensor are kind of annoyingly complicated. Simply setting each one to 0 doesn't really give equations that are easy to solve. Hobson et al. come to the rescue: they suggest to multiply R_{11} by $\frac{B}{A}$ and then add it to R_{00} . Actually, the thing to do is the following:

$$rB(r)[B(r)R_{00} + A(r)R_{11}] = A'B + AB' = (AB)' = 0. \quad (7.18)$$

Therefore,

$$B = \frac{\alpha}{A}, \quad (7.19)$$

for some constant α . Plugging this into the equation $R_{33} = 0$ implies the equation

$$A + rA' = \alpha \quad \implies \quad (rA)' = \alpha. \quad (7.20)$$

The solution is

$$A(r) = \alpha \left(1 + \frac{k}{r}\right), \quad B(r) = \left(1 + \frac{k}{r}\right)^{-1}, \quad (7.21)$$

where k is an integration constant.

The Newtonian limit of g_{00} is

$$\alpha \left(1 + \frac{k}{r}\right) \xrightarrow{\text{Newt.}} 1 + 2\Phi. \quad (7.22)$$

Therefore,

$$\alpha = 1. \quad (7.23)$$

We already know Φ for a spherical mass distribution of total mass M :

$$\Phi = -\frac{GM}{r}. \quad (7.24)$$

Therefore,

$$k = -2GM. \quad (7.25)$$

Hence, the Schwarzschild metric,

$$ds^2 = \left(1 - \frac{2GM}{r}\right) dt^2 - \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (7.26)$$

In fact, *Birkhoff's theorem* states that the Schwarzschild metric is the unique spacetime geometry outside a spherically symmetric matter distribution. This is pretty surprising because why should the metric *have* to be static? The mass distribution does *not* have to be time-independent, as long as it is spherically symmetric! Nevertheless, the geometry outside the mass distribution would still be time-independent!

For example, if a spherically symmetric star were to pulsate purely radially for whatever reason, it would nevertheless not propagate any disturbance into the surrounding space! In particular, no gravitational waves!

The converse is not true, by the way. You can generate Schwarzschild by a non-spherically symmetric mass distribution (see 2304.12640)!

7.3 Gravitational Redshift

Consider an emitter at fixed spatial coordinates (r_E, θ_E, ϕ_E) and a receiver also at fixed coordinates (r_R, θ_R, ϕ_R) . Let t_E be the coordinate time of emission and t_R the receipt time. The path of the photon is null geodesic, $ds^2 = 0$. Define

$$\mu = GM, \quad (7.27)$$

then

$$\left(1 - \frac{2\mu}{r}\right) dt^2 = \left(1 - \frac{2\mu}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (7.28)$$

Thus, with λ an affine parameter,

$$\frac{dt}{d\lambda} = \left(1 - \frac{2\mu}{r}\right)^{-1/2} \left(-g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}\right)^{1/2}, \quad (7.29)$$

or

$$t_R - t_E = \int_{\lambda_E}^{\lambda_R} d\lambda \left(1 - \frac{2\mu}{r}\right)^{-1/2} \left(-g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}\right)^{1/2}. \quad (7.30)$$

The RHS above depends only on the path through space. So, if the emitter and receiver have fixed spatial coordinates, then $t_R - t_E$ is the same for *any* photon signal sent by the emitter at any time. In particular, the difference in coordinate time between any two emissions is exactly the same as the difference in coordinate time between the corresponding two receipts:

$$\Delta t_R = \Delta t_E. \quad (7.31)$$

That's *coordinate* time. The proper time experience by the emitter and the receiver, however, are different. Since their spatial positions are fixed, for each of them, $dr = d\theta = d\phi = 0$, and so the Schwarzschild line element becomes

$$d\tau^2 = ds^2 = \left(1 - \frac{2\mu}{r}\right) dt^2. \quad (7.32)$$

Thus, the proper time experienced by each is

$$\Delta\tau_E = \left(1 - \frac{2\mu}{r_E}\right)^{1/2} \Delta t_E, \quad \Delta\tau_R = \left(1 - \frac{2\mu}{r_R}\right)^{1/2} \Delta t_R, \quad (7.33)$$

and, since $\Delta t_E = \Delta t_R$, the ratio is

$$\frac{\Delta\tau_R}{\Delta\tau_E} = \left(\frac{1 - \frac{2\mu}{r_R}}{1 - \frac{2\mu}{r_E}}\right)^{1/2}. \quad (7.34)$$

Suppose the receiver is further away from the central mass distribution ($r_R > r_E$). Then, $\Delta\tau_R > \Delta\tau_E$. In other words, time passes more slowly for the emitter, who is closer to the mass distribution where the gravitational field is stronger. This is gravitational redshift.

The name redshift is motivated by thinking about what happens to the photon in the signal itself. Suppose that $\Delta\tau_E$ coincides with the period of the photon as observed by the emitter. Then the frequency, ν_E , as observed by the emitter is the inverse of that. Similarly, the frequency observed by the receiver is the inverse of $\Delta\tau_R$. Thus,

$$\frac{\nu_R}{\nu_E} = \left(\frac{1 - \frac{2\mu}{r_E}}{1 - \frac{2\mu}{r_R}}\right)^{1/2}. \quad (7.35)$$

So, if $r_R > r_E$, then $\nu_R < \nu_E$, that is the frequency of the light observed by the receiver is lower (i.e., more “red”) than the frequency of the light observed by the emitter. The redshift factor z is defined such that

$$1 + z = \frac{\nu_E}{\nu_R}. \quad (7.36)$$

7.4 Geodesics

The geodesic equation reads

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} = 0. \quad (7.37)$$

These become the equations (9.17) – (9.20). If we plug in $\theta = \pi/2$ for “equatorial plane” then we get (9.21) – (9.23).

Here k is the total energy for a particle of unit rest mass, $k = E/m$. On the other hand, h is negative of the angular momentum per unit mass, aka the specific angular momentum.

The r -component of the geodesic equation is rather complicated. It is easier to integrate it once. Then, this condition splits into two: one for *non-null* geodesic and one for *null* geodesics:

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 1, \quad (7.38a)$$

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0. \quad (7.38b)$$

7.5 Trajectories of Massive Particles

Again, let us choose the equatorial plane $\theta = \pi/2$. For a massive particle, the trajectory must be time-like. Thus, we can choose the affine paramter to just be the proper time τ . Then,

$$\left(1 - \frac{2\mu}{r}\right) \dot{t} = k, \quad (7.39a)$$

$$\left(1 - \frac{2\mu}{r}\right) \dot{t}^2 - \left(1 - \frac{2\mu}{r}\right)^{-1} \dot{r}^2 - r^2 \dot{\phi}^2 = 1, \quad (7.39b)$$

$$r^2 \dot{\phi} = h. \quad (7.39c)$$

Plugging in the first and last equations into the second one gives

$$\boxed{\dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{r}\right) - \frac{2GM}{r} = k^2 - 1.} \quad (7.40)$$

Note that $h = r^2 \dot{\phi}$ implies

$$\frac{dr}{d\tau} = \frac{dr}{d\phi} \frac{d\phi}{d\tau} = \frac{h}{r^2} \frac{dr}{d\phi}, \quad (7.41)$$

and so,

$$\left(\frac{h}{r^2} \frac{dr}{d\phi}\right)^2 + \frac{h^2}{r^2} = k^2 - 1 + \frac{2GM}{r} + \frac{2GMh^2}{r^3}. \quad (7.42)$$

Let

$$u = \frac{1}{r}, \quad (7.43)$$

so that

$$\left(\frac{du}{d\phi}\right)^2 + u^2 = \frac{k^2 - 1}{h^2} + \frac{2GMu}{h^2} + 2GMu^3. \quad (7.44)$$

Differentiate this with respect to ϕ :

$$\boxed{\frac{d^2 u}{d\phi^2} + u = \frac{GM}{h^2} + 3GMu^2.} \quad (7.45)$$

Compare to Newtonian limit:

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) + \frac{GMm}{r}. \quad (7.46)$$

Equations of motion:

$$r^2\dot{\phi} = h, \quad \ddot{r} = \frac{h}{r^3} - \frac{GM}{r^2}. \quad (7.47)$$

Then,

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{h^2}, \quad (7.48)$$

which looks almost the same! Just beware that the r in the Newtonian equation is the radial *distance*, whereas the r that is implicit in (7.45) is the radial *coordinate*, which is related to a radial distance via the metric.

7.6 Radial Motion of Massive Particles

For radial motion, $\dot{\pi} = 0$ and so $h = r^2\dot{\phi} = 0$. We'd have to be very careful if we wanted to use (7.48) in this case because the whole equation would have to blow up. Instead, let's use (7.47):

$$\ddot{r} = -\frac{GM}{r^2}. \quad (7.49)$$

This looks deceptively identical to the Newtonian equation for radial acceleration in the presence of a Newtonian gravitation force. However, we have to remember two things:

1. In (7.49), r is a coordinate, the the radial distance – distances, of course, will involve the metric;
2. In (7.49), the time derivative is with respect to *proper* time.

Let's instead study the integrated version of the above equation, which is just (7.40) with $h = 0$:

$$\dot{r}^2 - \frac{2GM}{r} = k^2 - 1. \quad (7.50)$$

Here, k is an integration constant. How do we determine k ? By initial conditions (or some other condition at some other time, e.g., final conditions). For example, if we say that an object is dropped from radius R initially at rest, then, at $\tau = 0$, we have $\dot{r} = 0$ and the above equation reads

$$-\frac{2GM}{R} = k^2 - 1 \quad \implies \quad k^2 = 1 - \frac{2GM}{R}, \quad (7.51)$$

and so

$$\frac{1}{2}\dot{r}^2 = GM\left(\frac{1}{r} - \frac{1}{R}\right). \quad (7.52)$$

Again, this looks deceptively the same as the Newtonian formula for the gain in kinetic energy due to the loss in gravitational potential energy as the object falls from radius R to radius r of a unit mass. However, you have to keep points 1 and 2 above in mind.

Suppose we take $R \rightarrow \infty$ so that the object starts at rest at infinite radius and falls in. In this case, (7.51) gives $k = 1$. Thus, the geodesic equations (7.39) simplify to

$$\frac{dt}{d\tau} = \left(1 - \frac{2\mu}{r}\right)^{-1}, \quad (7.53a)$$

$$\frac{dr}{d\tau} = -\left(\frac{2\mu}{r}\right)^{1/2}. \quad (7.53b)$$

The sign of the RHS of the last equation is ambiguous because we have to take a square root. The sign is chosen to be radially *infalling*. The 4-velocity is

$$u^\mu = \frac{dx^\mu}{d\tau} = \left(\left(1 - \frac{2\mu}{r}\right)^{-1}, -\left(\frac{2\mu}{r}\right)^{1/2}, 0, 0 \right). \quad (7.54)$$

First, we can solve for r as a function of τ since the equation is separable:

$$\frac{2}{3} \frac{d}{d\tau} r^{3/2} = \sqrt{r} \frac{dr}{d\tau} = -\sqrt{2\mu} \quad \implies \quad \frac{d}{d\tau} r^{3/2} = -\frac{3}{2} \sqrt{2\mu}. \quad (7.55)$$

So,

$$r^{3/2}(\tau) = -\frac{3}{2} \sqrt{2\mu} \tau + c, \quad (7.56)$$

where c is a constant. Let $r(\tau = 0) = r_0$, so that $c = r_0^{3/2}$, and

$$r^{3/2}(\tau) = r_0^{3/2} - \frac{3}{2} \sqrt{2\mu} \tau \quad \implies \quad \boxed{r(\tau) = r_0 \left(1 - \frac{3\sqrt{2\mu}}{2r_0^{3/2}} \tau\right)^{2/3}}. \quad (7.57)$$

Hobson et al. solves for τ in terms of r instead:

$$\boxed{\tau(r) = \frac{2}{3\sqrt{2\mu}} (r_0^{3/2} - r^{3/2})}. \quad (7.58)$$

Now, we combine the two equations to write

$$\frac{dr}{dt} = -\left(1 - \frac{2\mu}{r}\right) \left(\frac{2\mu}{r}\right)^{1/2}. \quad (7.59)$$

Let

$$u = \left(\frac{2\mu}{r}\right)^{1/2} \quad \implies \quad r = \frac{2\mu}{u^2} \quad \implies \quad dr = -\frac{4\mu}{u^3} du, \quad (7.60)$$

and

$$-\frac{4\mu}{u^3} \frac{du}{dt} = -(1 - u^2)u \quad \implies \quad \frac{1}{u^4(1 - u^2)} \frac{du}{dt} = \frac{1}{4\mu}. \quad (7.61)$$

By partial fractions decomposition,

$$\frac{1}{u^4(1 - u^2)} = \frac{1}{u^4} + \frac{1}{u^2} + \frac{1}{2(1 + u)} + \frac{1}{2(1 - u)}, \quad (7.62)$$

and so

$$\frac{1}{4\mu} = \left(\frac{1}{u^4} + \frac{1}{u^2} + \frac{1}{2(1 + u)} + \frac{1}{2(1 - u)} \right) \frac{du}{dt} = \frac{d}{dt} \left(-\frac{1}{3u^3} - \frac{1}{u} + \frac{1}{2} \ln(1 + u) - \frac{1}{2} \ln(1 - u) \right). \quad (7.63)$$

We may write this as

$$\frac{d}{dt} \left(-\frac{4\mu}{3u^3} - \frac{4\mu}{u} + 2\mu \ln \frac{\frac{1}{u} + 1}{\frac{1}{u} - 1} \right) = 1. \quad (7.64)$$

Returning to r :

$$\frac{d}{dt} \left(-\frac{2}{3} \sqrt{\frac{r^3}{2\mu}} - 4\mu \sqrt{\frac{r}{2\mu}} + 2\mu \ln \frac{\sqrt{\frac{r}{2\mu}} + 1}{\sqrt{\frac{r}{2\mu}} - 1} \right) = 1. \quad (7.65)$$

The solution is

$$t(r) = \frac{2}{3} \left(\sqrt{\frac{r_0^3}{2\mu}} - \sqrt{\frac{r^3}{2\mu}} \right) + 4\mu \left(\sqrt{\frac{r_0}{2\mu}} - \sqrt{\frac{r}{2\mu}} \right) + 2\mu \ln \left(\frac{\sqrt{\frac{r}{2\mu}} + 1}{\sqrt{\frac{r}{2\mu}} - 1} \frac{\sqrt{\frac{r_0}{2\mu}} - 1}{\sqrt{\frac{r_0}{2\mu}} + 1} \right). \quad (7.66)$$

Note that

$$\tau(0) = \frac{2}{3} \sqrt{\frac{r_0^3}{2\mu}}, \quad (7.67)$$

and so the proper time experienced by an infalling observer in falling all the way from r_0 to the singularity at $r = 0$ is finite!

On the other hand,

$$t(r = 2\mu) = \infty. \quad (7.68)$$

The coordinate time is the proper time experienced by stationary observer far away at large radius. Therefore, from the perspective of such an observer, it looks like it takes an infalling particle infinite time to reach the horizon at $r = 2\mu$!

Suppose you have a stationary observer at radius r and an infalling particle that passes by. What velocity does the stationary observer measure for the particle at the moment it passes by? Well, the relationship between a proper time interval dt' as measured by the stationary observer and a proper time interval dt is

$$dt' = \left(1 - \frac{2\mu}{r} \right)^{1/2} dt. \quad (7.69)$$

Similarly, the radial distance measured by the stationary observer is related to the radial coordinate separation dr via

$$dr' = \left(1 - \frac{2\mu}{r} \right)^{-1/2} dr. \quad (7.70)$$

Thus, the stationary observer measures the velocity of the passing infalling particle to be

$$\frac{dr'}{dt'} = \left(1 - \frac{2\mu}{r} \right)^{-1} \frac{dr}{dt} = - \left(\frac{2\mu}{r} \right)^{1/2}. \quad (7.71)$$

This approaches 1 as $r \rightarrow 2\mu$. That is, from the perspective of a stationary observer hovering close to the horizon, the infalling particle passes by at basically the speed of light!

This is only valid for $r > 2\mu$, which is good since, otherwise, the formula would give a speed faster than that of light for $r < 2\mu$. The physical reason why the above formula only holds for $r > 2\mu$ is that it is actually not possible for an observer to be stationary once inside the horizon.

Chapter 8

Experimental Tests of GR

8.1 Precession of Planetary Orbits

The shape equation for the geodesic path of a massive particle in the Schwarzschild background is eq. (9.33) of Hobson et al.:

$$\frac{d^2u}{d\phi^2} + u - \frac{GM}{h^2} = 3GMu^2. \quad (8.1)$$

The Newtonina version of this is eq. (9.34):

$$\frac{d^2u}{d\phi^2} + u - \frac{GM}{h^2} = 0. \quad (8.2)$$

Clearly, if the Newtonian limit is supposed to be any good at describing planetary orbits, it must be the case that the term $3GMu^2$ appearing on the RHS of the GR result versus the Newtonian result is “small.” But, at the moment, it’s dimensionful, so “small” makes no sense. We should make it dimensionless. Since we have set the speed of light equal to 1, length and time are the same units. Since $h = r^2\dot{\phi}$, it follows that h has units of length. So, hu is dimensionless. Thus, define

$$\xi = hu. \quad (8.3)$$

Then, the shape equation reads

$$\frac{d^2\xi}{d\phi^2} + \xi - \frac{GM}{h} = \frac{3GM}{h}\xi^2. \quad (8.4)$$

The quantity

$$\gamma = \frac{GM}{h} \quad (8.5)$$

is dimensionless. So, our equation is

$$\frac{d^2\xi}{d\phi^2} + \xi - \gamma = 3\gamma\xi^2. \quad (8.6)$$

In the Newtonian limit, the RHS is 0 and you can see that the solution for ξ is of order γ . Therefore, it makes sense to define the parameter

$$\lambda = 3\gamma^2, \quad (8.7)$$

so that the equation reads

$$\frac{d^2\xi}{d\phi^2} + \xi - \gamma = \lambda \left(\frac{\xi}{\gamma} \right)^2. \quad (8.8)$$

This way, we know that the RHS is $O(\lambda)$, since ξ/γ is order 1. So, now we can assert that

$$\lambda \ll 1. \quad (8.9)$$

Let's verify that this is true for Mercury going around the Sun. The average orbital radius of Mercury is 5.79×10^{10} m and its orbital period is 88 days or 7.6×10^6 s. The angular speed is

$$\dot{\phi} = \frac{2\pi}{7.6 \times 10^6 \text{ s}} = 8.3 \times 10^{-7} \text{ s}^{-1}, \quad (8.10)$$

and the angular momentum per unit mass is

$$h = r^2 \dot{\phi} = (5.79 \times 10^{10} \text{ m})^2 (8.3 \times 10^{-7} \text{ s}^{-1}) = 2.8 \times 10^{15} \text{ m}^2/\text{s}. \quad (8.11)$$

The mass of the Sun is 2×10^{30} kg and so

$$\frac{GM}{c} = \frac{(6.67 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2})(2 \times 10^{30} \text{ kg})}{3 \times 10^8 \text{ m/s}} = 4.45 \times 10^{11} \text{ m}^2/\text{s}, \quad (8.12)$$

and

$$\gamma = \frac{GM}{hc} = \frac{4.45 \times 10^{11} \text{ m}^2/\text{s}}{2.8 \times 10^{15} \text{ m}^2/\text{s}} = 1.6 \times 10^{-4}. \quad (8.13)$$

Note that we have to divide GM/h by c in order to make it dimensionless. Finally,

$$\lambda = 3\gamma^3 = 1.24 \times 10^{-11} \ll 1. \quad (8.14)$$

So, this suggests solving the problem perturbatively: order-by-order in λ . First, let's solve the equation in the limit $\lambda = 0$ (i.e., the Newtonian limit),

$$\xi_0(\phi) = \gamma(1 + e \cos \phi). \quad (8.15)$$

Sure enough, you can verify for yourself that this satisfies the equation

$$\frac{d^2 \xi}{d\phi^2} + \xi - \gamma = 0. \quad (8.16)$$

Now, let's set

$$\xi = \xi_0 + \lambda \xi_1, \quad (8.17)$$

and keep only the terms in the equation that are linear-order in λ :

$$\frac{d^2 \xi_1}{d\phi^2} + \xi_1 = (1 + e \cos \phi)^2. \quad (8.18)$$

This isn't the easiest thing to solve, but you can use `DSolve` in Mathematica to find the most general solution. A specific solution is given in Hobson et al. eq. (10.4):

$$\xi_1 = 1 + e^2 \left(\frac{1}{2} - \frac{1}{6} \cos 2\phi \right) + e\phi \sin \phi. \quad (8.19)$$

As ϕ increases (i.e., the planet keeps orbiting over and over again through many periods, only the last term continues to grow while the other terms are either constant or bounded. Therefore, since the coefficient λ of this term is so small, let us only keep the last term above, which is the *cumulative* term (i.e., the one that accumulates through many orbits). Hence,

$$\xi = \gamma(1 + e \cos \phi) + \lambda e \phi \sin \phi. \quad (8.20)$$

Define

$$\alpha = \frac{\lambda}{\gamma} = 3\gamma^2, \quad (8.21)$$

so that

$$\xi = \gamma[1 + e(\cos \phi + \alpha \phi \sin \phi)]. \quad (8.22)$$

Finally,

$$\cos[(1 - \alpha)\phi] = \cos(\alpha\phi) \cos \phi + \sin(\alpha\phi) \sin \phi \approx \cos \phi + \alpha \phi \sin \phi, \quad (8.23)$$

where we used the smallness of α . Hence,

$$\xi = \gamma\{1 + e \cos[(1 - \alpha)\phi]\}. \quad (8.24)$$

Make sure you understand how this is equivalent to eq. (10.7) of Hobson et al. The orbit is periodic, but the period is not 2π anymore, but rather $\frac{2\pi}{1-\alpha}$, which is a little bit bigger by the amount

$$\Delta\phi = \frac{2\pi}{1-\alpha} - 2\pi \approx 2\pi\alpha = 6\pi \left(\frac{GM}{hc}\right)^2. \quad (8.25)$$

So far, we've been using h calculated using the *average* radius of orbit of Mercury as though its orbit were a circle. Of course, its orbit is actually an ellipse with some semimajor axis a and eccentricity e . The parameters a , e , and h are related via eq. (10.2) of Hobson et al.:

$$a = \frac{h^2}{GM(1-e^2)}, \quad (8.26)$$

which you can get from Kepler's first law of planetary motion. Thus,

$$\Delta\phi = \frac{6\pi GM}{a(1-e^2)c^2}. \quad (8.27)$$

For the actual elliptical orbit of Mercury, $a = 5.8 \times 10^{10}$ m and $e = 0.2$. Thus,

$$\Delta\phi = 5.01 \times 10^{-7} \text{ rad/orbit} = 0.103''/\text{orbit}, \quad (8.28)$$

where '' stands for arcsecond: $1/3600$ of a degree. If you use a circular orbit instead and just use h in (8.11) directly, you get 4.8×10^{-7} rad/orbit instead, which is not bad. This is reported in Hobson et al. in the usual convention of per century rather than per orbit:

$$\Delta\phi = \frac{0.103''}{\text{orbit}} \frac{\text{orbit}}{88 \text{ days}} \frac{1000 \text{ days}}{\text{century}} = 43.0''/\text{century}. \quad (8.29)$$

8.2 Bending of Light

Here is a quick and dirty way to get the shape equation for a photon trajectory from the massive case: just send $h \rightarrow \infty$. This is because h is the angular momentum per unit mass. In the same way that you might have naively thought that a photon can't have momentum because it has zero mass, you might have thought that it also can't have angular momentum. But, you learn in relativity that a photon does actually have momentum and, for the same reason, it also has angular momentum. But, then, that suggests that h , which is the angular momentum per unit mass should go to infinity if the mass goes to 0. If we do that, we get

$$\frac{d^2u}{d\phi^2} + u = 3GMu^2. \quad (8.30)$$

We can't use $\xi = hu$ as our dimensionless quantity anymore because, we just said that $h \rightarrow \infty$ for a photon. We need some other unit of length. The relevant one here is the *impact parameter*, b , which is the transverse distance between the initial far-away trajectory of the photon and the point mass which is deflecting it. So, we just replace h with b :

$$\xi = bu, \quad \lambda = \frac{3GM}{b}, \quad (8.31)$$

so that

$$\frac{d^2\xi}{d\phi^2} + \xi = \lambda\xi^2. \quad (8.32)$$

Why is it that in the massive case, (8.8), I divided ξ by γ on the RHS and defined $\lambda \sim (GM/h)^3$ accordingly, whereas in the massless case, I don't divide ξ by anything on the RHS and so $\lambda \sim GM/b$? The reason is that, in the massive case, when $\lambda = 0$, the solution for ξ is $O(\gamma)$. In the massless case, however, when $\lambda = 0$, the solution for ξ is $O(1)$: the solution to

$$\frac{d^2\xi}{d\phi^2} + \xi = 0, \quad (8.33)$$

is simply

$$\xi_0 = \sin \phi. \quad (8.34)$$

Of course, this isn't the most general solution to the equation. It is the physically sensible one: $\xi = bu = \frac{b}{r}$ and so this solution says $b = r \sin \phi$. Well, $r \sin \phi$ is just the vertical distance between the trajectory at the point mass. This just says that that is constant and equal to the impact parameter b .

Again, we set

$$\xi = \xi_0 + \lambda\xi_1, \quad (8.35)$$

and keep only up to $O(\lambda)$:

$$\frac{d^2\xi_1}{d\phi^2} + \xi_1 = \sin^2 \phi, \quad (8.36)$$

a solution to which is

$$\xi_1 = \frac{1}{2}(1 + \frac{1}{3} \cos 2\phi). \quad (8.37)$$

Thus,

$$\xi = \sin \phi + \frac{1}{2}\lambda(1 + \frac{1}{3} \cos 2\phi). \quad (8.38)$$

When $\lambda = 0$ and the path is just straight, the incoming path is the limit $\phi \rightarrow \pi$, and the outgoing path is the limit $\phi \rightarrow 0$. So, expand the above equation around $\phi = \pi$:

$$\xi \xrightarrow{\phi \rightarrow \pi} \frac{2\lambda}{3} - (\phi - \pi). \quad (8.39)$$

We are looking for where this vanishes because $\xi \rightarrow 0$ as $r \rightarrow \infty$:

$$\phi_{\text{in}} = \pi + \frac{2\lambda}{3}. \quad (8.40)$$

Similarly, we expand around $\phi = 0$:

$$\xi \xrightarrow{\phi \rightarrow 0} \phi + \frac{2\lambda}{3}, \quad (8.41)$$

which vanishes at

$$\phi_{\text{out}} = -\frac{2\lambda}{3}. \quad (8.42)$$

Therefore, the deflection angle is

$$\Delta\phi = \phi_{\text{in}} - \phi_{\text{out}} - \pi = \frac{4\lambda}{3} = \frac{4GM}{c^2 b}. \quad (8.43)$$

Again, I've put c^2 back in to get the dimensions right.

Plug in the mass of the Sun and the radius of the Sun for b (the photon grazes the surface of the Sun¹) to get

$$\Delta\phi = 1.75'', \quad (8.44)$$

which was corroborated by Eddington's 1919 solar eclipse expedition.

¹Note that the closest approach distance of the deflected path is strictly smaller than the impact parameter b . Therefore, technically, this light ray would crash into the Sun! However, the deflection is so tiny in the first place that the difference between the radius of the Sun and the actual impact parameter needed in order to just graze the surface of the Sun is a higher-order effect and can be neglected.